Discrete Krylov representation of the 3p vertex (`vertex_krylov`)

This notebook is the complete record of results of the `vertex_krylov` task: Step 1 (phases V0, V1, V2, V3a, V3b) and Step 2 (V3c). It implements and tests Algorithm 2 of the project note **[VD]** *Jacobi matrices and couplings of the 3p vertex from discrete spectral data*, which takes discrete 2p and 3p PSF weights on one symmetric grid and returns the Jacobi matrices $\bar{\mathbb{J}}$, $\bar{N}$ ($\bar{\mathbb{J}}_d = -\bar{\mathbb{J}}^T$), $\mathbb{J}_B$, N_B, $\mathbb{J}_{B^\dagger}$, $N_{B^\dagger}$, the couplings (VD27) and the self-energy coefficients Σ^{L}, Σ^{R} (VD4). Inserted into (VD6)–(VD12) they give the Matsubara regular part $\tilde{K}^{(12)}$ of Lihm's 3p estimator. [VD] Theorem 2 claims exactness for exact Lehmann data at $T > 0$ and for binned data with bins symmetric about zero.

regime	data source	exactness	section
Hubbard atom (V1)	HubbardAtom ED, ExactGrid	exact	2
short Wilson chain (V2a/V2b)	mpNRG ED of the chain, ExactGrid / LogGrid	exact	3
same chain, untruncated NRG (V3a)	mpNRG NRG, LogGrid	exact	4
decoupled tail, truncated NRG (V3b)	mpNRG NRG, LogGrid	exact	5
physical chain, truncated NRG (V3c)	mpNRG NRG, LogGrid	**not** exact; measured (report only)	6

Criteria: the task sheet as amended after the Step-1 review (amendments A1–A3 in `vertex_krylov_STATUS.md`).

Code. `src/vertex_krylov.jl` (package `HubbardAtom`, imported as `HA`; depends on `LinearAlgebra` only) and `mpNRG/test/vertex_krylov_adapter.jl` (data from mpNRG, case runners). The tests `test/test_vertex_krylov.jl` (V0, V1) and `mpNRG/test/test_vertex_krylov_nrg.jl` (V2, V3a, V3b) assert the same checks with the same functions.

Conventions. $H_{\mathrm{int}} = U n_\uparrow n_\downarrow$; $q_\sigma = [d_\sigma, H_{\mathrm{int}}]$, $q_\sigma^\dagger = [H_{\mathrm{int}}, d_\sigma^\dagger]$, $B = q_{i_1 i_2} = \{q_{i_1}, d_{i_2}^\dagger\}$ (called `Bq` in code; `siam(B = …)` is the magnetic field). Channel (↑↓): $(i_1, i_2) = (\downarrow, \downarrow)$, particle chain for $\sigma = \uparrow$; channel (↑↑): $(i_1, i_2) = (\uparrow, \uparrow)$. Rows $X = (d_\sigma^\dagger, q_\sigma^\dagger)$, $X_d = (d_\sigma, q_\sigma)$. PSFs follow Kugler Eq. (28), without permutation signs.

In [1]:

```
using Pkg
Pkg.activate(normpath(joinpath(@__DIR__, "..", "mpNRG", "notebooks")); io = devnull)   # the notebook environment
using MultipointNRG, LinearAlgebra, Printf
import HubbardAtom as HA
using CairoMakie
CairoMakie.activate!(type = "png")
const ROOT = normpath(joinpath(@__DIR__, ".."))
include(joinpath(ROOT, "mpNRG", "test", "helpers.jl"))
include(joinpath(ROOT, "mpNRG", "test", "vertex_krylov_adapter.jl"))
FIG = joinpath(@__DIR__, "figures", "vertex_krylov"); mkpath(FIG)
savefig(name, fig) = (save(joinpath(FIG, name * ".png"), fig; px_per_unit = 2); fig)
fl(x) = max(x, 1e-17)
println("Julia ", VERSION, ", threads ", Threads.nthreads(), ", BLAS threads ", BLAS.get_num_threads())
```

```
Julia 1
.13.3, threads 4, BLAS threads 2
```

Computation of all cases

All cases of the Step-1 test matrix are computed here, once, by the case runners of the adapter (`vk_run` , `vk_case_ed` , `vk_case_nrg`), which call `HA.vertex_krylov` and `HA.vk_checks` . The sections below only display and plot these results. Timings per case are printed and collected in `TIMES` .

In [2]:

```
R = Dict{String,Any}()      # all results, keyed by case label
# VK_NB_SMOKE=1: structural test run of this notebook on the Nch = 1 data only (the Nch = 2 labels then
# point to the Nch = 1 results). The committed notebook is executed without it.
const SMOKE = get(ENV, "VK_NB_SMOKE", "0") == "1"
NCHS = SMOKE ? (1,) : (1, 2)
TIMES = Dict{String,Any}()
tstart = time()

# --- V1: Hubbard atom, U = 1, εd = -0.3 ------------------------------------------------
const U_AT, ED_AT = 1.0, -0.3
function atom_case(T, chan; σ = :up, h = 0.0, i12 = nothing, rot = nothing)
    m = HA.HubbardAtomModel(U_AT, 1 / T)
    sp = HA.atom_spectrum(m; εd = ED_AT, h = h)
    t0 = time()
    d = HA.discrete_data_atom(m, sp; σ = σ, channel = chan, εd = ED_AT, h = h, i12 = i12, rot = rot)
    tm = Dict{Symbol,Float64}(:data => time() - t0)
    out = vk_run(d; bigfloat = d.N == 1, timings = tm)
    out[:data] = d
    d.N == 1 && rot === nothing && (out[:closed] = HA.atom_closed_form(out[:res]))
    return out
end
for T in (0.1, 0.5), chan in (:updn, :upup)
    R["V1 atom T=$T $chan"] = atom_case(T, chan)
end
for h in (0.0, 0.07)
    R["V1 atom N=2 h=$h"] = atom_case(0.1, :updn; σ = [:up, :dn], i12 = (:dn, :dn), h = h)
    R["V1 atom N=1 up h=$h"] = atom_case(0.1, :updn; σ = :up, h = h)
    R["V1 atom N=1 dn h=$h"] = atom_case(0.1, :upup; σ = :dn, h = h)
end
θ, φ = 0.4, 0.9
ROT = [cos(θ) exp(im * φ) * sin(θ); -exp(-im * φ) * sin(θ) cos(θ)]
R["V1 atom N=2 rotated"] = atom_case(0.1, :updn; σ = [:up, :dn], i12 = (1, 1), h = 0.07, rot = ROT)
@printf("V1 done: %.1f s\n", time() - tstart)

# --- V2: Wilson chain, ED (exact and binned) -------------------------------------------
KEEP = Set(["V2 Nch=2 B=0.0 updn exact", "V2 Nch=2 B=0.0 updn log", "V2 Nch=1 B=0.0 updn exact", "V3a Nch=2 B=0.0 updn"])
SMOKE && (KEEP = union(KEEP, Set(replace.(collect(KEEP), "Nch=2" => "Nch=1"))))
for Nch in NCHS, field in (0.0, 0.05), chan in (:updn, :upup), gk in (:exact, :log)
```

```
    t0 = time()
    k = "V2 Nch=$Nch B=$field $chan $gk"
    # BigFloat particle chain only for Nch ≤ 1 (Nch = 2: R ≈ 2600 rows, L ≈ 2600 steps; deviation in STATUS)
    R[k] = vk_case_ed(Nch, gk, chan; field = field, bigfloat = gk === :exact && Nch == 1, keep = k in KEEP)
    GC.gc()
    @printf("V2 Nch=%d B=%.2f %-5s %-5s: %6.1f s\n", Nch, field, chan, gk, time() - t0)
end
t0 = time()
R["V2 block N=2"] = vk_case_ed(1, :exact, :updn; field = 0.05, σ = [:up, :dn], i12 = (:dn, :dn))
R["V2 block up"] = vk_case_ed(1, :exact, :updn; field = 0.05)
R["V2 block dn"] = vk_case_ed(1, :exact, :upup; field = 0.05, σ = :dn)
@printf("V2 block test: %.1f s\n", time() - t0)
# smallest reproducing case of finding F1 (sites -1..0, 16 states)
R["V2 Nch=0 B=0.0 updn exact"] = vk_case_ed(0, :exact, :updn; bigfloat = true)

# --- V3a: untruncated NRG ------------------------------------------------------------
for Nch in NCHS, field in (0.0, 0.05), chan in (:updn, :upup)
    t0 = time()
    k = "V3a Nch=$Nch B=$field $chan"
    R[k] = vk_case_nrg(Nch, chan; field = field, keep = k in KEEP)
    GC.gc()
    @printf("V3a Nch=%d B=%.2f %-5s: %6.1f s\n", Nch, field, chan, time() - t0)
end

# --- V3b: decoupled tail, truncated NRG ----------------------------------------------
for chan in (:updn, :upup)
    t0 = time()
    chs = decoupled_tail(wilson_chains(vk_model(), 3), 1)
    R["V3b $chan"] = vk_case_nrg(3, chan; Nkeep = 40, chains = chs)
    @printf("V3b %-5s: %6.1f s\n", chan, time() - t0)
end

# --- V3c: physical chain, truncated NRG (report only) --------------------------------
# NRG data are built by mpNRG/scripts/vertex_krylov_v3c.jl (vk_v3c_data) and cached in
# notebooks/data/vertex_krylov_v3c/; missing cases are built here. Algorithm and checks run here.
using JLD2
include(joinpath(ROOT, "mpNRG", "scripts", "vertex_krylov_v3c.jl"))   # V3C_TS, V3C_NKEEPS, v3c_file, v3c_build_all
v3c_build_all()
for T in V3C_TS, Nk in V3C_NKEEPS, chan in (:updn, :upup)
    t0 = time()
    d = load(v3c_file(T, Nk, chan), "data")
    o = vk_case_v3c(d)
    o[:data] = d
    R["V3c T=$T Nk=$Nk $chan"] = o
    @printf("V3c T=%.0e Nkeep=%3d %-5s: %6.1f s (cached NRG data: %.0f s)\n", T, Nk, chan, time() - t0, d.meta[:time_nrg] + d.meta[:time_data])
end
if SMOKE
    for k in collect(keys(R))
        occursin("Nch=1", k) && (R[replace(k, "Nch=1" => "Nch=2")] = R[k])
    end
end
println("SMOKE = ", SMOKE)
@printf("all cases: %.1f min\n", (time() - tstart) / 60)
```

```
V1 done: 54.3 s
V2 Nch=1 B=0.00 updn  exact:   22.7 s
V2 Nch=1 B=0.00 updn  log  :    2.1 s
V2 Nch=1 B=0.00 upup  exact:   13.1 s
V2 Nch=1 B=0.00 upup  log  :    1.2 s
V2 Nch=1 B=0.05 updn  exact:   27.3 s
V2 Nch=1 B=0.05 updn  log  :    1.3 s
V2 Nch=1 B=0.05 upup  exact:   27.4 s
V2 Nch=1 B=0.05 upup  log  :    1.3 s
V2 Nch=2 B=0.00 updn  exact:  529.0 s
V2 Nch=2 B=0.00 updn  log  :    6.1 s
V2 Nch=2 B=0.00 upup  exact:  537.7 s
V2 Nch=2 B=0.00 upup  log  :    6.2 s
V2 Nch=2 B=0.05 updn  exact: 1508.3 s
V2 Nch=2 B=0.05 updn  log  :    6.5 s
V2 Nch=2 B=0.05 upup  exact: 1543.5 s
V2 Nch=2 B=0.05 upup  log  :    6.5 s
V2 block test: 26.6 s
V3a Nch=1 B=0.00 updn :   33.6 s
```

```
V3a Nch=1 B=0.00 upup :    2.4 s
V3a Nch=1 B=0.05 updn :    2.7 s
V3a Nch=1 B=0.05 upup :    2.7 s
V3a Nch=2 B=0.00 updn :   11.8 s
V3a Nch=2 B=0.00 upup :   11.5 s
V3a Nch=2 B=0.05 updn :   12.4 s
V3a Nch=2 B=0.05 upup :   12.6 s
V3b updn :    4.0 s
V3b upup :    3.0 s
V3c T=1e-02 Nkeep= 64 updn :   80.2 s (cached NRG data: 41 s)
V3c T=1e-02 Nkeep= 64 upup :   72.4 s (cached NRG data: 13 s)
┌ Warning: Assignment to `t0` in soft scope is ambiguous because a global variable by the same name exists: `t0` will be treated as a new local. Disambiguate by using `local t0` to suppress this warning or `global t0
└ @ /home/user/Multipoint_Correlation_Ftns/mpNRG/scripts/vertex_krylov_v3c.jl:41
V3c T=1e-02 Nkeep=128 updn :   97.5 s (cached NRG data: 38 s)
V3c T=1e-02 Nkeep=128 upup :   96.9 s (cached NRG data: 38 s)
V3c T=1e-02 Nkeep=256 updn :  113.7 s (cached NRG data: 171 s)
V3c T=1e-02 Nkeep=256 upup :  126.5 s (cached NRG data: 170 s)
V3c T=1e-02 Nkeep=512 updn :  137.2 s (cached NRG data: 686 s)
V3c T=1e-02 Nkeep=512 upup :  134.3 s (cached NRG data: 695 s)
V3c T=1e-04 Nkeep= 64 updn :  132.7 s (cached NRG data: 27 s)
V3c T=1e-04 Nkeep= 64 upup :  139.1 s (cached NRG data: 28 s)
V3c T=1e-04 Nkeep=128 updn :  160.3 s (cached NRG data: 74 s)
V3c T=1e-04 Nkeep=128 upup :  171.2 s (cached NRG data: 75 s)
V3c T=1e-04 Nkeep=256 updn :  186.7 s (cached NRG data: 386 s)
V3c T=1e-04 Nkeep=256 upup :  176.4 s (cached NRG data: 381 s)
V3c T=1e-04 Nkeep=512 updn :  234.0 s (cached NRG data: 1506 s)
V3c T=1e-04 Nkeep=512 upup :  217.8 s (cached NRG data: 1530 s)
SMOKE = false
all cases: 111.7 min
```

0. Summary

The full test matrix of Steps 1 and 2 (task sheet Sec. 6): phase × case × channel × check, with the achieved value, the tolerance, and pass / FAIL / report. **Guaranteed** tolerances follow from [VD] Theorem 2 (stop rule); "report" means measured only. Generated from `R` by the same check functions as the tests.

- C1 self-energy (algebraic), C2 Zacinskis cross-check, C3 Dyson, C4 bath leakage, C5 hole symmetry, C6 Gauss quadrature, C7 factorization residual, C8 assembly.
- **Amended criteria** (task-sheet amendment of `vertex_krylov_STATUS.md`): (A1) for exact data the guaranteed C4 is that of the stabilized particle chain of [VD] Prop. 13 (projector $1 - \Omega\Omega^\dagger$); C4 of the unstabilized chain ($1 - vv^\dagger$) is report. (A2) For exact data the stabilized chain is the default output of Stage 2; the unstabilized chain is kept as a diagnostic. (A3) C2: the invariants (Σ^{S}, spectra and first-block weights of Ω', $S = \bar{N}^2$, start block) are guaranteed at 10^{-10}; the Jacobi-block agreement (`n_agree` , finding F2) is report.
- V3c (physical chain, truncated NRG) is report only, every check.
- C8 is relative to $\max|K^{\mathrm{direct}}|$, except where $K^{\mathrm{direct}} \equiv 0$ (marked `†` , see deviation 4 of `vertex_krylov_STATUS.md`).

In [3]:
```
ROWS = Any[]
function row!(phase, case, chan, check, val, tol; note = "")
    st = tol === nothing ? "report" : isnan(val) ? "n/a" : (val ≤ tol ? "pass" : "FAIL")
    push!(ROWS, (phase = phase, case = case, chan = chan, check = check, value = val, tol = tol, status = st, note = note))
end
TOL = Dict("V1" => (C1 = 1e-11, C2 = 1e-10, C34 = 1e-11, C56 = 1e-12, C7 = 1e-11, C8 = 1e-10),
```

```
            "V2a" => (C1 = 1e-10, C2 = 1e-10, C34 = 1e-9, C56 = 1e-12, C7 = 1e-10, C8 = 1e-9),
            "V2b" => (C1 = 1e-10, C2 = 1e-10, C34 = nothing, C56 = 1e-12, C7 = 1e-10, C8 = 1e-9),
            "V3a" => (C1 = 1e-10, C2 = 1e-10, C34 = nothing, C56 = 1e-11, C7 = 1e-9, C8 = 1e-8),
            "V3b" => (C1 = 1e-10, C2 = 1e-10, C34 = nothing, C56 = 1e-11, C7 = 1e-9, C8 = 1e-8),
            "V3c" => (C1 = nothing, C2 = nothing, C34 = nothing, C56 = nothing, C7 = nothing, C8 = nothing))
function rows_for!(phase, case, chan, out)
    t = TOL[phase]; c = out[:checks]
    row!(phase, case, chan, "C1", c[:C1], t.C1)
    row!(phase, case, chan, "C2", c[:C2], t.C2)
    haskey(c, :zac) && row!(phase, case, chan, "C2 blocks n_agree/L", c[:zac].n_agree / c[:zac].L_chain, nothing)
    row!(phase, case, chan, "C3", c[:C3], t.C34)
    if haskey(out, :unstab)          # exact data: the default chain is the stabilized one (A1, A2)
        row!(phase, case, chan, "C4 (stabilized, default)", c[:C4], t.C34)
        row!(phase, case, chan, "C4 (unstabilized, diagnostic)", out[:unstab].C4, nothing)
    else
        row!(phase, case, chan, "C4", c[:C4], t.C34)
    end
    haskey(out, :big) && row!(phase, case, chan, "C4 (BigFloat, default chain)", out[:big].C4, t.C34)
    haskey(out, :big_unstab) && row!(phase, case, chan, "C4 (BigFloat, unstabilized)", out[:big_unstab].C4, nothing)
    row!(phase, case, chan, "C5", c[:C5], t.C56)
    row!(phase, case, chan, "C6", c[:C6], t.C56)
    row!(phase, case, chan, "C7", c[:C7], t.C7)
    row!(phase, case, chan, "C8", c[:C8], t.C8; note = c[:C8_parts].degenerate ? "†" : "")
end
chs(s) = s === :updn ? "(↑↓)" : "(↑↑)"
for T in (0.1, 0.5), chan in (:updn, :upup)
    rows_for!("V1", "atom T=$T", chs(chan), R["V1 atom T=$T $chan"])
    cf = R["V1 atom T=$T $chan"][:closed]
    row!("V1", "atom T=$T", chs(chan), "closed form |ΔJ|,|ΔÑ²|,|Δσ|", max(cf.J_err, cf.N2_err, cf.σ_err), 1e-12)
    row!("V1", "atom T=$T", chs(chan), "chain length 1, resid", cf.L == 1 ? cf.resid : Inf, 1e-12)
end
for h in (0.0, 0.07)
    rows_for!("V1", "atom N=2 h=$h", "B=q↓↓", R["V1 atom N=2 h=$h"])
end
rows_for!("V1", "atom N=2 rotated", "B=q′₁₁", R["V1 atom N=2 rotated"])
for Nch in (1, 2), field in (0.0, 0.05), chan in (:updn, :upup)
    case = "chain Nch=$Nch B=$field"
    rows_for!("V2a", case, chs(chan), R["V2 Nch=$Nch B=$field $chan exact"])
end
for Nch in (1, 2), field in (0.0, 0.05), chan in (:updn, :upup)
    rows_for!("V2b", "chain Nch=$Nch B=$field", chs(chan), R["V2 Nch=$Nch B=$field $chan log"])
end
rows_for!("V2a", "N=2 block, Nch=1 B=0.05", "B=q↓↓", R["V2 block N=2"])
for Nch in (1, 2), field in (0.0, 0.05), chan in (:updn, :upup)
    o = R["V3a Nch=$Nch B=$field $chan"]
    rows_for!("V3a", "NRG Nch=$Nch B=$field", chs(chan), o)
    row!("V3a", "NRG Nch=$Nch B=$field", chs(chan), "data vs V2b max|ΔW̃|,|Δs|", max(o[:diff].W, o[:diff].wB, o[:diff].s), nothing)
end
for chan in (:updn, :upup)
    o = R["V3b $chan"]
    rows_for!("V3b", "decoupled tail Nkeep=40", chs(chan), o)
    row!("V3b", "decoupled tail Nkeep=40", chs(chan), "data vs ED max|ΔW̃|,|Δs|", max(o[:diff].W, o[:diff].wB, o[:diff].s), nothing)
end
for T in V3C_TS, Nk in V3C_NKEEPS, chan in (:updn, :upup)
    o = R["V3c T=$T Nk=$Nk $chan"]; case = @sprintf("T=%.0e Nkeep=%d", T, Nk)
    rows_for!("V3c", case, chs(chan), o)
    ref = R["V3c T=$T Nk=512 $chan"][:checks][:box]; bx = o[:checks][:box]; m_ = .!isnan.(real.(bx.Kt))
    row!("V3c", case, chs(chan), "K̃ vs Nkeep=512 (rel.)", maximum(abs, bx.Kt[m_] - ref.Kt[m_]) / maximum(abs, ref.Kt[m_]), nothing)
end
@printf("%-5s %-26s %-7s %-30s %11s %9s  %s\n", "phase", "case", "channel", "check", "value", "tol", "status")
for r in ROWS
    @printf("%-5s %-26s %-7s %-30s %11.2e %9s  %s%s\n", r.phase, r.case, r.chan, r.check, r.value,
            r.tol === nothing ? "—" : @sprintf("%.0e", r.tol), r.status, r.note)
end
FAILS = filter(r -> r.status == "FAIL", ROWS)
println("\n", length(ROWS), " rows: ", count(r -> r.status == "pass", ROWS), " pass, ", length(FAILS), " FAIL, ",
        count(r -> r.status == "report", ROWS), " report")
```

```
phase case                       channel check                          value       tol  status
V1    atom T=0.1                 (↑↓)    C1                          7.98e-16     1e-11  pass
```

V1	atom T=0.1	(↑↓)	C2	2.83e-14	1e-10	pass
V1	atom T=0.1	(↑↓)	C2 blocks n_agree/L	1.00e+00	–	report
V1	atom T=0.1	(↑↓)	C3	1.68e-15	1e-11	pass
V1	atom T=0.1	(↑↓)	C4 (stabilized, default)	2.22e-16	1e-11	pass
V1	atom T=0.1	(↑↓)	C4 (unstabilized, diagnostic)	2.22e-16	–	report
V1	atom T=0.1	(↑↓)	C4 (BigFloat, default chain)	1.46e-16	1e-11	pass
V1	atom T=0.1	(↑↓)	C4 (BigFloat, unstabilized)	1.46e-16	–	report
V1	atom T=0.1	(↑↓)	C5	0.00e+00	1e-12	pass
V1	atom T=0.1	(↑↓)	C6	1.05e-16	1e-12	pass
V1	atom T=0.1	(↑↓)	C7	3.42e-16	1e-11	pass
V1	atom T=0.1	(↑↓)	C8	3.83e-16	1e-10	pass
V1	atom T=0.1	(↑↓)	closed form |ΔJ̃|,|ΔÑ²|,|Δσ|	8.33e-17	1e-12	pass
V1	atom T=0.1	(↑↓)	chain length 1, resid	1.73e-32	1e-12	pass
V1	atom T=0.1	(↑↑)	C1	7.98e-16	1e-11	pass
V1	atom T=0.1	(↑↑)	C2	2.83e-14	1e-10	pass
V1	atom T=0.1	(↑↑)	C2 blocks n_agree/L	1.00e+00	–	report
V1	atom T=0.1	(↑↑)	C3	1.68e-15	1e-11	pass
V1	atom T=0.1	(↑↑)	C4 (stabilized, default)	2.22e-16	1e-11	pass
V1	atom T=0.1	(↑↑)	C4 (unstabilized, diagnostic)	2.22e-16	–	report
V1	atom T=0.1	(↑↑)	C4 (BigFloat, default chain)	1.46e-16	1e-11	pass
V1	atom T=0.1	(↑↑)	C4 (BigFloat, unstabilized)	1.46e-16	–	report
V1	atom T=0.1	(↑↑)	C5	0.00e+00	1e-12	pass
V1	atom T=0.1	(↑↑)	C6	1.05e-16	1e-12	pass
V1	atom T=0.1	(↑↑)	C7	3.66e-16	1e-11	pass
V1	atom T=0.1	(↑↑)	C8	3.17e-16	1e-10	pass†
V1	atom T=0.1	(↑↑)	closed form |ΔJ̃|,|ΔÑ²|,|Δσ|	8.33e-17	1e-12	pass
V1	atom T=0.1	(↑↑)	chain length 1, resid	1.73e-32	1e-12	pass
V1	atom T=0.5	(↑↓)	C1	5.47e-16	1e-11	pass
V1	atom T=0.5	(↑↓)	C2	5.41e-15	1e-10	pass
V1	atom T=0.5	(↑↓)	C2 blocks n_agree/L	1.00e+00	–	report
V1	atom T=0.5	(↑↓)	C3	5.66e-15	1e-11	pass
V1	atom T=0.5	(↑↓)	C4 (stabilized, default)	1.12e-16	1e-11	pass
V1	atom T=0.5	(↑↓)	C4 (unstabilized, diagnostic)	1.12e-16	–	report
V1	atom T=0.5	(↑↓)	C4 (BigFloat, default chain)	1.67e-16	1e-11	pass
V1	atom T=0.5	(↑↓)	C4 (BigFloat, unstabilized)	1.67e-16	–	report
V1	atom T=0.5	(↑↓)	C5	0.00e+00	1e-12	pass
V1	atom T=0.5	(↑↓)	C6	1.15e-16	1e-12	pass
V1	atom T=0.5	(↑↓)	C7	1.57e-16	1e-11	pass
V1	atom T=0.5	(↑↓)	C8	1.45e-16	1e-10	pass
V1	atom T=0.5	(↑↓)	closed form |ΔJ̃|,|ΔÑ²|,|Δσ|	1.11e-16	1e-12	pass
V1	atom T=0.5	(↑↓)	chain length 1, resid	4.72e-34	1e-12	pass
V1	atom T=0.5	(↑↑)	C1	5.47e-16	1e-11	pass
V1	atom T=0.5	(↑↑)	C2	5.41e-15	1e-10	pass
V1	atom T=0.5	(↑↑)	C2 blocks n_agree/L	1.00e+00	–	report
V1	atom T=0.5	(↑↑)	C3	5.66e-15	1e-11	pass
V1	atom T=0.5	(↑↑)	C4 (stabilized, default)	1.12e-16	1e-11	pass
V1	atom T=0.5	(↑↑)	C4 (unstabilized, diagnostic)	1.12e-16	–	report
V1	atom T=0.5	(↑↑)	C4 (BigFloat, default chain)	1.67e-16	1e-11	pass
V1	atom T=0.5	(↑↑)	C4 (BigFloat, unstabilized)	1.67e-16	–	report
V1	atom T=0.5	(↑↑)	C5	0.00e+00	1e-12	pass
V1	atom T=0.5	(↑↑)	C6	1.15e-16	1e-12	pass
V1	atom T=0.5	(↑↑)	C7	1.57e-16	1e-11	pass
V1	atom T=0.5	(↑↑)	C8	1.46e-16	1e-10	pass†
V1	atom T=0.5	(↑↑)	closed form |ΔJ̃|,|ΔÑ²|,|Δσ|	1.11e-16	1e-12	pass
V1	atom T=0.5	(↑↑)	chain length 1, resid	4.72e-34	1e-12	pass
V1	atom N=2 h=0.0	B=q↓↓	C1	9.19e-16	1e-11	pass
V1	atom N=2 h=0.0	B=q↓↓	C2	2.26e-14	1e-10	pass
V1	atom N=2 h=0.0	B=q↓↓	C2 blocks n_agree/L	1.00e+00	–	report
V1	atom N=2 h=0.0	B=q↓↓	C3	1.68e-15	1e-11	pass
V1	atom N=2 h=0.0	B=q↓↓	C4 (stabilized, default)	0.00e+00	1e-11	pass
V1	atom N=2 h=0.0	B=q↓↓	C4 (unstabilized, diagnostic)	0.00e+00	–	report
V1	atom N=2 h=0.0	B=q↓↓	C5	0.00e+00	1e-12	pass
V1	atom N=2 h=0.0	B=q↓↓	C6	1.05e-16	1e-12	pass
V1	atom N=2 h=0.0	B=q↓↓	C7	7.31e-16	1e-11	pass
V1	atom N=2 h=0.0	B=q↓↓	C8	1.95e-16	1e-10	pass
V1	atom N=2 h=0.07	B=q↓↓	C1	1.82e-15	1e-11	pass
V1	atom N=2 h=0.07	B=q↓↓	C2	2.81e-14	1e-10	pass
V1	atom N=2 h=0.07	B=q↓↓	C2 blocks n_agree/L	1.00e+00	–	report
V1	atom N=2 h=0.07	B=q↓↓	C3	1.98e-15	1e-11	pass
V1	atom N=2 h=0.07	B=q↓↓	C4 (stabilized, default)	3.50e-16	1e-11	pass
V1	atom N=2 h=0.07	B=q↓↓	C4 (unstabilized, diagnostic)	3.50e-16	–	report
V1	atom N=2 h=0.07	B=q↓↓	C5	0.00e+00	1e-12	pass
V1	atom N=2 h=0.07	B=q↓↓	C6	1.98e-16	1e-12	pass
V1	atom N=2 h=0.07	B=q↓↓	C7	5.10e-16	1e-11	pass
V1	atom N=2 h=0.07	B=q↓↓	C8	8.30e-16	1e-10	pass
V1	atom N=2 rotated	B=q′₁₁	C1	1.08e-15	1e-11	pass
V1	atom N=2 rotated	B=q′₁₁	C2	3.37e-14	1e-10	pass
V1	atom N=2 rotated	B=q′₁₁	C2 blocks n_agree/L	1.00e+00	–	report
V1	atom N=2 rotated	B=q′₁₁	C3	2.82e-15	1e-11	pass

V1	atom N=2 rotated	$B=q'_{11}$	C4 (stabilized, default)	2.66e-16	1e-11	pass
V1	atom N=2 rotated	$B=q'_{11}$	C4 (unstabilized, diagnostic)	2.66e-16	—	report
V1	atom N=2 rotated	$B=q'_{11}$	C5	0.00e+00	1e-12	pass
V1	atom N=2 rotated	$B=q'_{11}$	C6	1.91e-16	1e-12	pass
V1	atom N=2 rotated	$B=q'_{11}$	C7	8.02e-16	1e-11	pass
V1	atom N=2 rotated	$B=q'_{11}$	C8	4.60e-16	1e-10	pass
V2a	chain Nch=1 B=0.0	(↑↓)	C1	2.87e-15	1e-10	pass
V2a	chain Nch=1 B=0.0	(↑↓)	C2	1.20e-13	1e-10	pass
V2a	chain Nch=1 B=0.0	(↑↓)	C2 blocks n_agree/L	4.48e-01	—	report
V2a	chain Nch=1 B=0.0	(↑↓)	C3	4.36e-14	1e-09	pass
V2a	chain Nch=1 B=0.0	(↑↓)	C4 (stabilized, default)	1.21e-14	1e-09	pass
V2a	chain Nch=1 B=0.0	(↑↓)	C4 (unstabilized, diagnostic)	8.45e-01	—	report
V2a	chain Nch=1 B=0.0	(↑↓)	C4 (BigFloat, default chain)	1.28e-14	1e-09	pass
V2a	chain Nch=1 B=0.0	(↑↓)	C4 (BigFloat, unstabilized)	8.35e-01	—	report
V2a	chain Nch=1 B=0.0	(↑↓)	C5	0.00e+00	1e-12	pass
V2a	chain Nch=1 B=0.0	(↑↓)	C6	7.55e-16	1e-12	pass
V2a	chain Nch=1 B=0.0	(↑↓)	C7	9.78e-16	1e-10	pass
V2a	chain Nch=1 B=0.0	(↑↓)	C8	4.66e-15	1e-09	pass
V2a	chain Nch=1 B=0.0	(↑↑)	C1	3.18e-15	1e-10	pass

V2a	chain	Nch=1 B=0.0	(↑↑)	C2	1.12e-13	1e-10	pass
V2a	chain	Nch=1 B=0.0	(↑↑)	C2 blocks n_agree/L	4.48e-01	–	report
V2a	chain	Nch=1 B=0.0	(↑↑)	C3	4.38e-14	1e-09	pass
V2a	chain	Nch=1 B=0.0	(↑↑)	C4 (stabilized, default)	1.20e-14	1e-09	pass
V2a	chain	Nch=1 B=0.0	(↑↑)	C4 (unstabilized, diagnostic)	8.30e-01	–	report
V2a	chain	Nch=1 B=0.0	(↑↑)	C4 (BigFloat, default chain)	1.28e-14	1e-09	pass
V2a	chain	Nch=1 B=0.0	(↑↑)	C4 (BigFloat, unstabilized)	8.14e-01	–	report
V2a	chain	Nch=1 B=0.0	(↑↑)	C5	0.00e+00	1e-12	pass
V2a	chain	Nch=1 B=0.0	(↑↑)	C6	7.68e-16	1e-12	pass
V2a	chain	Nch=1 B=0.0	(↑↑)	C7	8.15e-16	1e-10	pass
V2a	chain	Nch=1 B=0.0	(↑↑)	C8	4.15e-14	1e-09	pass
V2a	chain	Nch=1 B=0.05	(↑↓)	C1	2.18e-15	1e-10	pass
V2a	chain	Nch=1 B=0.05	(↑↓)	C2	1.52e-13	1e-10	pass
V2a	chain	Nch=1 B=0.05	(↑↓)	C2 blocks n_agree/L	4.41e-01	–	report
V2a	chain	Nch=1 B=0.05	(↑↓)	C3	6.54e-14	1e-09	pass
V2a	chain	Nch=1 B=0.05	(↑↓)	C4 (stabilized, default)	4.99e-16	1e-09	pass
V2a	chain	Nch=1 B=0.05	(↑↓)	C4 (unstabilized, diagnostic)	4.74e-01	–	report
V2a	chain	Nch=1 B=0.05	(↑↓)	C4 (BigFloat, default chain)	8.19e-16	1e-09	pass
V2a	chain	Nch=1 B=0.05	(↑↓)	C4 (BigFloat, unstabilized)	5.56e-01	–	report
V2a	chain	Nch=1 B=0.05	(↑↓)	C5	0.00e+00	1e-12	pass
V2a	chain	Nch=1 B=0.05	(↑↓)	C6	1.22e-15	1e-12	pass
V2a	chain	Nch=1 B=0.05	(↑↓)	C7	8.72e-16	1e-10	pass
V2a	chain	Nch=1 B=0.05	(↑↓)	C8	3.24e-15	1e-09	pass
V2a	chain	Nch=1 B=0.05	(↑↑)	C1	1.35e-15	1e-10	pass
V2a	chain	Nch=1 B=0.05	(↑↑)	C2	1.51e-13	1e-10	pass
V2a	chain	Nch=1 B=0.05	(↑↑)	C2 blocks n_agree/L	4.41e-01	–	report
V2a	chain	Nch=1 B=0.05	(↑↑)	C3	6.54e-14	1e-09	pass
V2a	chain	Nch=1 B=0.05	(↑↑)	C4 (stabilized, default)	4.99e-16	1e-09	pass
V2a	chain	Nch=1 B=0.05	(↑↑)	C4 (unstabilized, diagnostic)	5.50e-01	–	report
V2a	chain	Nch=1 B=0.05	(↑↑)	C4 (BigFloat, default chain)	8.12e-16	1e-09	pass
V2a	chain	Nch=1 B=0.05	(↑↑)	C4 (BigFloat, unstabilized)	5.55e-01	–	report
V2a	chain	Nch=1 B=0.05	(↑↑)	C5	0.00e+00	1e-12	pass
V2a	chain	Nch=1 B=0.05	(↑↑)	C6	9.66e-16	1e-12	pass
V2a	chain	Nch=1 B=0.05	(↑↑)	C7	8.36e-16	1e-10	pass
V2a	chain	Nch=1 B=0.05	(↑↑)	C8	2.37e-14	1e-09	pass
V2a	chain	Nch=2 B=0.0	(↑↓)	C1	4.41e-15	1e-10	pass
V2a	chain	Nch=2 B=0.0	(↑↓)	C2	1.60e-13	1e-10	pass
V2a	chain	Nch=2 B=0.0	(↑↓)	C2 blocks n_agree/L	8.00e-02	–	report
V2a	chain	Nch=2 B=0.0	(↑↓)	C3	1.05e-13	1e-09	pass
V2a	chain	Nch=2 B=0.0	(↑↓)	C4 (stabilized, default)	4.88e-15	1e-09	pass
V2a	chain	Nch=2 B=0.0	(↑↓)	C4 (unstabilized, diagnostic)	2.48e-01	–	report
V2a	chain	Nch=2 B=0.0	(↑↓)	C5	0.00e+00	1e-12	pass
V2a	chain	Nch=2 B=0.0	(↑↓)	C6	2.74e-15	1e-12	pass
V2a	chain	Nch=2 B=0.0	(↑↓)	C7	9.72e-16	1e-10	pass
V2a	chain	Nch=2 B=0.0	(↑↓)	C8	4.78e-14	1e-09	pass
V2a	chain	Nch=2 B=0.0	(↑↑)	C1	4.74e-15	1e-10	pass
V2a	chain	Nch=2 B=0.0	(↑↑)	C2	1.22e-12	1e-10	pass
V2a	chain	Nch=2 B=0.0	(↑↑)	C2 blocks n_agree/L	8.04e-02	–	report
V2a	chain	Nch=2 B=0.0	(↑↑)	C3	1.05e-13	1e-09	pass
V2a	chain	Nch=2 B=0.0	(↑↑)	C4 (stabilized, default)	4.88e-15	1e-09	pass
V2a	chain	Nch=2 B=0.0	(↑↑)	C4 (unstabilized, diagnostic)	2.36e-01	–	report
V2a	chain	Nch=2 B=0.0	(↑↑)	C5	0.00e+00	1e-12	pass
V2a	chain	Nch=2 B=0.0	(↑↑)	C6	2.44e-15	1e-12	pass
V2a	chain	Nch=2 B=0.0	(↑↑)	C7	4.97e-16	1e-10	pass
V2a	chain	Nch=2 B=0.0	(↑↑)	C8	2.80e-13	1e-09	pass
V2a	chain	Nch=2 B=0.05	(↑↓)	C1	6.72e-15	1e-10	pass
V2a	chain	Nch=2 B=0.05	(↑↓)	C2	2.42e-13	1e-10	pass
V2a	chain	Nch=2 B=0.05	(↑↓)	C2 blocks n_agree/L	2.75e-01	–	report
V2a	chain	Nch=2 B=0.05	(↑↓)	C3	1.60e-13	1e-09	pass
V2a	chain	Nch=2 B=0.05	(↑↓)	C4 (stabilized, default)	5.57e-15	1e-09	pass
V2a	chain	Nch=2 B=0.05	(↑↓)	C4 (unstabilized, diagnostic)	1.99e-01	–	report
V2a	chain	Nch=2 B=0.05	(↑↓)	C5	0.00e+00	1e-12	pass
V2a	chain	Nch=2 B=0.05	(↑↓)	C6	5.38e-15	1e-12	pass
V2a	chain	Nch=2 B=0.05	(↑↓)	C7	5.56e-16	1e-10	pass
V2a	chain	Nch=2 B=0.05	(↑↓)	C8	8.16e-14	1e-09	pass
V2a	chain	Nch=2 B=0.05	(↑↑)	C1	6.01e-15	1e-10	pass
V2a	chain	Nch=2 B=0.05	(↑↑)	C2	1.95e-13	1e-10	pass
V2a	chain	Nch=2 B=0.05	(↑↑)	C2 blocks n_agree/L	2.72e-01	–	report
V2a	chain	Nch=2 B=0.05	(↑↑)	C3	1.60e-13	1e-09	pass
V2a	chain	Nch=2 B=0.05	(↑↑)	C4 (stabilized, default)	5.57e-15	1e-09	pass
V2a	chain	Nch=2 B=0.05	(↑↑)	C4 (unstabilized, diagnostic)	2.01e-01	–	report
V2a	chain	Nch=2 B=0.05	(↑↑)	C5	0.00e+00	1e-12	pass
V2a	chain	Nch=2 B=0.05	(↑↑)	C6	5.55e-15	1e-12	pass
V2a	chain	Nch=2 B=0.05	(↑↑)	C7	6.63e-16	1e-10	pass
V2a	chain	Nch=2 B=0.05	(↑↑)	C8	7.37e-14	1e-09	pass
V2b	chain	Nch=1 B=0.0	(↑↓)	C1	1.93e-15	1e-10	pass
V2b	chain	Nch=1 B=0.0	(↑↓)	C2	5.73e-14	1e-10	pass
V2b	chain	Nch=1 B=0.0	(↑↓)	C2 blocks n_agree/L	3.26e-01	–	report
V2b	chain	Nch=1 B=0.0	(↑↓)	C3	2.03e-01	–	report

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V2b   chain Nch=1 B=0.0          (↑↓)    C4                              3.72e-01        –  report
V2b   chain Nch=1 B=0.0          (↑↓)    C5                              0.00e+00     1e-12  pass
V2b   chain Nch=1 B=0.0          (↑↓)    C6                              2.87e-16     1e-12  pass
V2b   chain Nch=1 B=0.0          (↑↓)    C7                              6.52e-16     1e-10  pass
V2b   chain Nch=1 B=0.0          (↑↓)    C8                              2.83e-15     1e-09  pass
V2b   chain Nch=1 B=0.0          (↑↑)    C1                              1.93e-15     1e-10  pass
V2b   chain Nch=1 B=0.0          (↑↑)    C2                              5.73e-14     1e-10  pass
V2b   chain Nch=1 B=0.0          (↑↑)    C2 blocks n_agree/L             3.26e-01        –  report
V2b   chain Nch=1 B=0.0          (↑↑)    C3                              2.03e-01        –  report
V2b   chain Nch=1 B=0.0          (↑↑)    C4                              3.72e-01        –  report
V2b   chain Nch=1 B=0.0          (↑↑)    C5                              0.00e+00     1e-12  pass
V2b   chain Nch=1 B=0.0          (↑↑)    C6                              7.72e-16     1e-12  pass
V2b   chain Nch=1 B=0.0          (↑↑)    C7                              6.52e-16     1e-10  pass
V2b   chain Nch=1 B=0.0          (↑↑)    C8                              2.20e-14     1e-09  pass
V2b   chain Nch=1 B=0.05         (↑↓)    C1                              1.76e-15     1e-10  pass
V2b   chain Nch=1 B=0.05         (↑↓)    C2                              5.64e-14     1e-10  pass
V2b   chain Nch=1 B=0.05         (↑↓)    C2 blocks n_agree/L             5.15e-01        –  report
V2b   chain Nch=1 B=0.05         (↑↓)    C3                              2.39e-01        –  report
V2b   chain Nch=1 B=0.05         (↑↓)    C4                              5.73e-01        –  report
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V2b	chain Nch=1 B=0.05	(↑↓)	C5	0.00e+00	1e-12	pass
V2b	chain Nch=1 B=0.05	(↑↓)	C6	8.95e-16	1e-12	pass
V2b	chain Nch=1 B=0.05	(↑↓)	C7	2.40e-14	1e-10	pass
V2b	chain Nch=1 B=0.05	(↑↓)	C8	2.75e-14	1e-09	pass
V2b	chain Nch=1 B=0.05	(↑↑)	C1	1.76e-15	1e-10	pass
V2b	chain Nch=1 B=0.05	(↑↑)	C2	5.64e-14	1e-10	pass
V2b	chain Nch=1 B=0.05	(↑↑)	C2 blocks n_agree/L	5.15e-01	–	report
V2b	chain Nch=1 B=0.05	(↑↑)	C3	2.39e-01	–	report
V2b	chain Nch=1 B=0.05	(↑↑)	C4	5.73e-01	–	report
V2b	chain Nch=1 B=0.05	(↑↑)	C5	0.00e+00	1e-12	pass
V2b	chain Nch=1 B=0.05	(↑↑)	C6	7.47e-16	1e-12	pass
V2b	chain Nch=1 B=0.05	(↑↑)	C7	3.45e-14	1e-10	pass
V2b	chain Nch=1 B=0.05	(↑↑)	C8	1.78e-13	1e-09	pass
V2b	chain Nch=2 B=0.0	(↑↓)	C1	2.36e-15	1e-10	pass
V2b	chain Nch=2 B=0.0	(↑↓)	C2	2.59e-14	1e-10	pass
V2b	chain Nch=2 B=0.0	(↑↓)	C2 blocks n_agree/L	2.04e-01	–	report
V2b	chain Nch=2 B=0.0	(↑↓)	C3	4.54e-01	–	report
V2b	chain Nch=2 B=0.0	(↑↓)	C4	3.76e-01	–	report
V2b	chain Nch=2 B=0.0	(↑↓)	C5	0.00e+00	1e-12	pass
V2b	chain Nch=2 B=0.0	(↑↓)	C6	3.88e-16	1e-12	pass
V2b	chain Nch=2 B=0.0	(↑↓)	C7	4.69e-16	1e-10	pass
V2b	chain Nch=2 B=0.0	(↑↓)	C8	1.95e-14	1e-09	pass
V2b	chain Nch=2 B=0.0	(↑↑)	C1	2.36e-15	1e-10	pass
V2b	chain Nch=2 B=0.0	(↑↑)	C2	2.59e-14	1e-10	pass
V2b	chain Nch=2 B=0.0	(↑↑)	C2 blocks n_agree/L	2.04e-01	–	report
V2b	chain Nch=2 B=0.0	(↑↑)	C3	4.54e-01	–	report
V2b	chain Nch=2 B=0.0	(↑↑)	C4	3.76e-01	–	report
V2b	chain Nch=2 B=0.0	(↑↑)	C5	0.00e+00	1e-12	pass
V2b	chain Nch=2 B=0.0	(↑↑)	C6	6.30e-16	1e-12	pass
V2b	chain Nch=2 B=0.0	(↑↑)	C7	6.20e-16	1e-10	pass
V2b	chain Nch=2 B=0.0	(↑↑)	C8	4.22e-14	1e-09	pass
V2b	chain Nch=2 B=0.05	(↑↓)	C1	2.05e-15	1e-10	pass
V2b	chain Nch=2 B=0.05	(↑↓)	C2	3.83e-14	1e-10	pass
V2b	chain Nch=2 B=0.05	(↑↓)	C2 blocks n_agree/L	1.84e-01	–	report
V2b	chain Nch=2 B=0.05	(↑↓)	C3	3.63e-01	–	report
V2b	chain Nch=2 B=0.05	(↑↓)	C4	3.38e-01	–	report
V2b	chain Nch=2 B=0.05	(↑↓)	C5	0.00e+00	1e-12	pass
V2b	chain Nch=2 B=0.05	(↑↓)	C6	7.16e-16	1e-12	pass
V2b	chain Nch=2 B=0.05	(↑↓)	C7	8.11e-16	1e-10	pass
V2b	chain Nch=2 B=0.05	(↑↓)	C8	2.72e-14	1e-09	pass
V2b	chain Nch=2 B=0.05	(↑↑)	C1	2.05e-15	1e-10	pass
V2b	chain Nch=2 B=0.05	(↑↑)	C2	3.83e-14	1e-10	pass
V2b	chain Nch=2 B=0.05	(↑↑)	C2 blocks n_agree/L	1.84e-01	–	report
V2b	chain Nch=2 B=0.05	(↑↑)	C3	3.63e-01	–	report
V2b	chain Nch=2 B=0.05	(↑↑)	C4	3.38e-01	–	report
V2b	chain Nch=2 B=0.05	(↑↑)	C5	0.00e+00	1e-12	pass
V2b	chain Nch=2 B=0.05	(↑↑)	C6	9.65e-16	1e-12	pass
V2b	chain Nch=2 B=0.05	(↑↑)	C7	7.95e-16	1e-10	pass
V2b	chain Nch=2 B=0.05	(↑↑)	C8	2.88e-14	1e-09	pass
V2a	N=2 block, Nch=1 B=0.05	B=q↓↓	C1	2.69e-15	1e-10	pass
V2a	N=2 block, Nch=1 B=0.05	B=q↓↓	C2	9.26e-14	1e-10	pass
V2a	N=2 block, Nch=1 B=0.05	B=q↓↓	C2 blocks n_agree/L	3.76e-01	–	report
V2a	N=2 block, Nch=1 B=0.05	B=q↓↓	C3	4.53e-14	1e-09	pass
V2a	N=2 block, Nch=1 B=0.05	B=q↓↓	C4 (stabilized, default)	1.13e-15	1e-09	pass
V2a	N=2 block, Nch=1 B=0.05	B=q↓↓	C4 (unstabilized, diagnostic)	6.44e-01	–	report
V2a	N=2 block, Nch=1 B=0.05	B=q↓↓	C5	0.00e+00	1e-12	pass
V2a	N=2 block, Nch=1 B=0.05	B=q↓↓	C6	7.11e-16	1e-12	pass
V2a	N=2 block, Nch=1 B=0.05	B=q↓↓	C7	7.91e-16	1e-10	pass
V2a	N=2 block, Nch=1 B=0.05	B=q↓↓	C8	8.74e-15	1e-09	pass
V3a	NRG Nch=1 B=0.0	(↑↓)	C1	2.03e-15	1e-10	pass
V3a	NRG Nch=1 B=0.0	(↑↓)	C2	6.97e-14	1e-10	pass
V3a	NRG Nch=1 B=0.0	(↑↓)	C2 blocks n_agree/L	3.26e-01	–	report
V3a	NRG Nch=1 B=0.0	(↑↓)	C3	2.03e-01	–	report
V3a	NRG Nch=1 B=0.0	(↑↓)	C4	3.72e-01	–	report
V3a	NRG Nch=1 B=0.0	(↑↓)	C5	0.00e+00	1e-11	pass
V3a	NRG Nch=1 B=0.0	(↑↓)	C6	5.17e-16	1e-11	pass
V3a	NRG Nch=1 B=0.0	(↑↓)	C7	5.83e-16	1e-09	pass
V3a	NRG Nch=1 B=0.0	(↑↓)	C8	3.12e-15	1e-08	pass
V3a	NRG Nch=1 B=0.0	(↑↓)	data vs V2b max|ΔW̃|,|Δs|	4.44e-15	–	report
V3a	NRG Nch=1 B=0.0	(↑↑)	C1	2.03e-15	1e-10	pass
V3a	NRG Nch=1 B=0.0	(↑↑)	C2	6.97e-14	1e-10	pass
V3a	NRG Nch=1 B=0.0	(↑↑)	C2 blocks n_agree/L	3.26e-01	–	report
V3a	NRG Nch=1 B=0.0	(↑↑)	C3	2.03e-01	–	report
V3a	NRG Nch=1 B=0.0	(↑↑)	C4	3.72e-01	–	report
V3a	NRG Nch=1 B=0.0	(↑↑)	C5	0.00e+00	1e-11	pass
V3a	NRG Nch=1 B=0.0	(↑↑)	C6	7.17e-16	1e-11	pass
V3a	NRG Nch=1 B=0.0	(↑↑)	C7	5.05e-16	1e-09	pass
V3a	NRG Nch=1 B=0.0	(↑↑)	C8	3.28e-14	1e-08	pass
V3a	NRG Nch=1 B=0.0	(↑↑)	data vs V2b max|ΔW̃|,|Δs|	4.44e-15	–	report

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V3a   NRG Nch=1 B=0.05          (↑↓)    C1                              2.56e-15     1e-10  pass
V3a   NRG Nch=1 B=0.05          (↑↓)    C2                              3.61e-14     1e-10  pass
V3a   NRG Nch=1 B=0.05          (↑↓)    C2 blocks n_agree/L             5.15e-01         –  report
V3a   NRG Nch=1 B=0.05          (↑↓)    C3                              2.39e-01         –  report
V3a   NRG Nch=1 B=0.05          (↑↓)    C4                              5.73e-01         –  report
V3a   NRG Nch=1 B=0.05          (↑↓)    C5                              0.00e+00     1e-11  pass
V3a   NRG Nch=1 B=0.05          (↑↓)    C6                              7.07e-16     1e-11  pass
V3a   NRG Nch=1 B=0.05          (↑↓)    C7                              2.40e-14     1e-09  pass
V3a   NRG Nch=1 B=0.05          (↑↓)    C8                              2.81e-14     1e-08  pass
V3a   NRG Nch=1 B=0.05          (↑↓)    data vs V2b max|ΔW̃|,|Δs|        1.67e-15         –  report
V3a   NRG Nch=1 B=0.05          (↑↑)    C1                              2.56e-15     1e-10  pass
V3a   NRG Nch=1 B=0.05          (↑↑)    C2                              3.61e-14     1e-10  pass
V3a   NRG Nch=1 B=0.05          (↑↑)    C2 blocks n_agree/L             5.15e-01         –  report
V3a   NRG Nch=1 B=0.05          (↑↑)    C3                              2.39e-01         –  report
V3a   NRG Nch=1 B=0.05          (↑↑)    C4                              5.73e-01         –  report
V3a   NRG Nch=1 B=0.05          (↑↑)    C5                              0.00e+00     1e-11  pass
V3a   NRG Nch=1 B=0.05          (↑↑)    C6                              7.15e-16     1e-11  pass
V3a   NRG Nch=1 B=0.05          (↑↑)    C7                              3.45e-14     1e-09  pass
V3a   NRG Nch=1 B=0.05          (↑↑)    C8                              1.76e-13     1e-08  pass
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V3a	NRG Nch=1 B=0.05	(↑↑)	data vs V2b max|ΔW̃|,|Δs|	1.67e-15	–	report
V3a	NRG Nch=2 B=0.0	(↑↓)	C1	1.68e-15	1e-10	pass
V3a	NRG Nch=2 B=0.0	(↑↓)	C2	3.21e-14	1e-10	pass
V3a	NRG Nch=2 B=0.0	(↑↓)	C2 blocks n_agree/L	2.04e-01	–	report
V3a	NRG Nch=2 B=0.0	(↑↓)	C3	4.54e-01	–	report
V3a	NRG Nch=2 B=0.0	(↑↓)	C4	3.76e-01	–	report
V3a	NRG Nch=2 B=0.0	(↑↓)	C5	0.00e+00	1e-11	pass
V3a	NRG Nch=2 B=0.0	(↑↓)	C6	4.40e-16	1e-11	pass
V3a	NRG Nch=2 B=0.0	(↑↓)	C7	5.71e-16	1e-09	pass
V3a	NRG Nch=2 B=0.0	(↑↓)	C8	2.13e-14	1e-08	pass
V3a	NRG Nch=2 B=0.0	(↑↓)	data vs V2b max|ΔW̃|,|Δs|	9.21e-15	–	report
V3a	NRG Nch=2 B=0.0	(↑↑)	C1	1.68e-15	1e-10	pass
V3a	NRG Nch=2 B=0.0	(↑↑)	C2	3.21e-14	1e-10	pass
V3a	NRG Nch=2 B=0.0	(↑↑)	C2 blocks n_agree/L	2.04e-01	–	report
V3a	NRG Nch=2 B=0.0	(↑↑)	C3	4.54e-01	–	report
V3a	NRG Nch=2 B=0.0	(↑↑)	C4	3.76e-01	–	report
V3a	NRG Nch=2 B=0.0	(↑↑)	C5	0.00e+00	1e-11	pass
V3a	NRG Nch=2 B=0.0	(↑↑)	C6	4.34e-16	1e-11	pass
V3a	NRG Nch=2 B=0.0	(↑↑)	C7	6.20e-16	1e-09	pass
V3a	NRG Nch=2 B=0.0	(↑↑)	C8	3.85e-14	1e-08	pass
V3a	NRG Nch=2 B=0.0	(↑↑)	data vs V2b max|ΔW̃|,|Δs|	9.21e-15	–	report
V3a	NRG Nch=2 B=0.05	(↑↓)	C1	1.79e-15	1e-10	pass
V3a	NRG Nch=2 B=0.05	(↑↓)	C2	5.21e-14	1e-10	pass
V3a	NRG Nch=2 B=0.05	(↑↓)	C2 blocks n_agree/L	1.84e-01	–	report
V3a	NRG Nch=2 B=0.05	(↑↓)	C3	3.63e-01	–	report
V3a	NRG Nch=2 B=0.05	(↑↓)	C4	3.38e-01	–	report
V3a	NRG Nch=2 B=0.05	(↑↓)	C5	0.00e+00	1e-11	pass
V3a	NRG Nch=2 B=0.05	(↑↓)	C6	6.48e-16	1e-11	pass
V3a	NRG Nch=2 B=0.05	(↑↓)	C7	4.98e-16	1e-09	pass
V3a	NRG Nch=2 B=0.05	(↑↓)	C8	2.71e-14	1e-08	pass
V3a	NRG Nch=2 B=0.05	(↑↓)	data vs V2b max|ΔW̃|,|Δs|	5.30e-15	–	report
V3a	NRG Nch=2 B=0.05	(↑↑)	C1	1.79e-15	1e-10	pass
V3a	NRG Nch=2 B=0.05	(↑↑)	C2	5.21e-14	1e-10	pass
V3a	NRG Nch=2 B=0.05	(↑↑)	C2 blocks n_agree/L	1.84e-01	–	report
V3a	NRG Nch=2 B=0.05	(↑↑)	C3	3.63e-01	–	report
V3a	NRG Nch=2 B=0.05	(↑↑)	C4	3.38e-01	–	report
V3a	NRG Nch=2 B=0.05	(↑↑)	C5	0.00e+00	1e-11	pass
V3a	NRG Nch=2 B=0.05	(↑↑)	C6	7.32e-16	1e-11	pass
V3a	NRG Nch=2 B=0.05	(↑↑)	C7	6.15e-16	1e-09	pass
V3a	NRG Nch=2 B=0.05	(↑↑)	C8	2.73e-14	1e-08	pass
V3a	NRG Nch=2 B=0.05	(↑↑)	data vs V2b max|ΔW̃|,|Δs|	5.30e-15	–	report
V3b	decoupled tail Nkeep=40	(↑↓)	C1	1.23e-15	1e-10	pass
V3b	decoupled tail Nkeep=40	(↑↓)	C2	7.12e-14	1e-10	pass
V3b	decoupled tail Nkeep=40	(↑↓)	C2 blocks n_agree/L	3.26e-01	–	report
V3b	decoupled tail Nkeep=40	(↑↓)	C3	2.03e-01	–	report
V3b	decoupled tail Nkeep=40	(↑↓)	C4	3.72e-01	–	report
V3b	decoupled tail Nkeep=40	(↑↓)	C5	0.00e+00	1e-11	pass
V3b	decoupled tail Nkeep=40	(↑↓)	C6	3.74e-16	1e-11	pass
V3b	decoupled tail Nkeep=40	(↑↓)	C7	4.89e-16	1e-09	pass
V3b	decoupled tail Nkeep=40	(↑↓)	C8	3.36e-15	1e-08	pass
V3b	decoupled tail Nkeep=40	(↑↓)	data vs ED max|ΔW̃|,|Δs|	3.22e-15	–	report
V3b	decoupled tail Nkeep=40	(↑↑)	C1	1.23e-15	1e-10	pass
V3b	decoupled tail Nkeep=40	(↑↑)	C2	7.12e-14	1e-10	pass
V3b	decoupled tail Nkeep=40	(↑↑)	C2 blocks n_agree/L	3.26e-01	–	report
V3b	decoupled tail Nkeep=40	(↑↑)	C3	2.03e-01	–	report
V3b	decoupled tail Nkeep=40	(↑↑)	C4	3.72e-01	–	report
V3b	decoupled tail Nkeep=40	(↑↑)	C5	0.00e+00	1e-11	pass
V3b	decoupled tail Nkeep=40	(↑↑)	C6	1.09e-15	1e-11	pass
V3b	decoupled tail Nkeep=40	(↑↑)	C7	9.89e-16	1e-09	pass
V3b	decoupled tail Nkeep=40	(↑↑)	C8	3.10e-14	1e-08	pass
V3b	decoupled tail Nkeep=40	(↑↑)	data vs ED max|ΔW̃|,|Δs|	3.22e-15	–	report
V3c	T=1e-02 Nkeep=64	(↑↓)	C1	1.28e-03	–	report
V3c	T=1e-02 Nkeep=64	(↑↓)	C2	6.45e-14	–	report
V3c	T=1e-02 Nkeep=64	(↑↓)	C2 blocks n_agree/L	8.16e-02	–	report
V3c	T=1e-02 Nkeep=64	(↑↓)	C3	5.65e-01	–	report
V3c	T=1e-02 Nkeep=64	(↑↓)	C4	4.19e-01	–	report
V3c	T=1e-02 Nkeep=64	(↑↓)	C4 (BigFloat, default chain)	4.19e-01	–	report
V3c	T=1e-02 Nkeep=64	(↑↓)	C5	0.00e+00	–	report
V3c	T=1e-02 Nkeep=64	(↑↓)	C6	2.26e-16	–	report
V3c	T=1e-02 Nkeep=64	(↑↓)	C7	1.40e-02	–	report
V3c	T=1e-02 Nkeep=64	(↑↓)	C8	1.16e-01	–	report
V3c	T=1e-02 Nkeep=64	(↑↓)	K̃ vs Nkeep=512 (rel.)	9.41e-02	–	report
V3c	T=1e-02 Nkeep=64	(↑↑)	C1	1.28e-03	–	report
V3c	T=1e-02 Nkeep=64	(↑↑)	C2	6.45e-14	–	report
V3c	T=1e-02 Nkeep=64	(↑↑)	C2 blocks n_agree/L	8.16e-02	–	report
V3c	T=1e-02 Nkeep=64	(↑↑)	C3	5.65e-01	–	report
V3c	T=1e-02 Nkeep=64	(↑↑)	C4	4.19e-01	–	report
V3c	T=1e-02 Nkeep=64	(↑↑)	C4 (BigFloat, default chain)	4.19e-01	–	report
V3c	T=1e-02 Nkeep=64	(↑↑)	C5	0.00e+00	–	report

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V3c   T=1e-02 Nkeep=64          (↑↑)   C6                              1.40e-15        –  report
V3c   T=1e-02 Nkeep=64          (↑↑)   C7                              2.06e-02        –  report
V3c   T=1e-02 Nkeep=64          (↑↑)   C8                              7.93e-02        –  report
V3c   T=1e-02 Nkeep=64          (↑↑)   K̃ vs Nkeep=512 (rel.)            5.91e-02        –  report
V3c   T=1e-02 Nkeep=128         (↑↓)   C1                              4.37e-04        –  report
V3c   T=1e-02 Nkeep=128         (↑↓)   C2                              3.02e-14        –  report
V3c   T=1e-02 Nkeep=128         (↑↓)   C2 blocks n_agree/L             7.76e-02        –  report
V3c   T=1e-02 Nkeep=128         (↑↓)   C3                              3.90e-01        –  report
V3c   T=1e-02 Nkeep=128         (↑↓)   C4                              4.30e-01        –  report
V3c   T=1e-02 Nkeep=128         (↑↓)   C4 (BigFloat, default chain)     4.30e-01        –  report
V3c   T=1e-02 Nkeep=128         (↑↓)   C5                              0.00e+00        –  report
V3c   T=1e-02 Nkeep=128         (↑↓)   C6                              2.02e-15        –  report
V3c   T=1e-02 Nkeep=128         (↑↓)   C7                              3.05e-03        –  report
V3c   T=1e-02 Nkeep=128         (↑↓)   C8                              5.37e-04        –  report
V3c   T=1e-02 Nkeep=128         (↑↓)   K̃ vs Nkeep=512 (rel.)            1.42e-02        –  report
V3c   T=1e-02 Nkeep=128         (↑↑)   C1                              4.37e-04        –  report
V3c   T=1e-02 Nkeep=128         (↑↑)   C2                              3.02e-14        –  report
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V3c	T=1e-02 Nkeep=128	(↑↑)	C2 blocks n_agree/L	7.76e-02	–	report
V3c	T=1e-02 Nkeep=128	(↑↑)	C3	3.90e-01	–	report
V3c	T=1e-02 Nkeep=128	(↑↑)	C4	4.30e-01	–	report
V3c	T=1e-02 Nkeep=128	(↑↑)	C4 (BigFloat, default chain)	4.30e-01	–	report
V3c	T=1e-02 Nkeep=128	(↑↑)	C5	0.00e+00	–	report
V3c	T=1e-02 Nkeep=128	(↑↑)	C6	1.81e-15	–	report
V3c	T=1e-02 Nkeep=128	(↑↑)	C7	2.28e-03	–	report
V3c	T=1e-02 Nkeep=128	(↑↑)	C8	4.62e-04	–	report
V3c	T=1e-02 Nkeep=128	(↑↑)	K̃ vs Nkeep=512 (rel.)	2.28e-02	–	report
V3c	T=1e-02 Nkeep=256	(↑↓)	C1	2.30e-06	–	report
V3c	T=1e-02 Nkeep=256	(↑↓)	C2	4.73e-13	–	report
V3c	T=1e-02 Nkeep=256	(↑↓)	C2 blocks n_agree/L	7.07e-02	–	report
V3c	T=1e-02 Nkeep=256	(↑↓)	C3	2.49e-01	–	report
V3c	T=1e-02 Nkeep=256	(↑↓)	C4	5.78e-01	–	report
V3c	T=1e-02 Nkeep=256	(↑↓)	C4 (BigFloat, default chain)	5.78e-01	–	report
V3c	T=1e-02 Nkeep=256	(↑↓)	C5	0.00e+00	–	report
V3c	T=1e-02 Nkeep=256	(↑↓)	C6	1.55e-15	–	report
V3c	T=1e-02 Nkeep=256	(↑↓)	C7	5.09e-08	–	report
V3c	T=1e-02 Nkeep=256	(↑↓)	C8	9.80e-05	–	report
V3c	T=1e-02 Nkeep=256	(↑↓)	K̃ vs Nkeep=512 (rel.)	5.54e-03	–	report
V3c	T=1e-02 Nkeep=256	(↑↑)	C1	2.30e-06	–	report
V3c	T=1e-02 Nkeep=256	(↑↑)	C2	4.73e-13	–	report
V3c	T=1e-02 Nkeep=256	(↑↑)	C2 blocks n_agree/L	7.07e-02	–	report
V3c	T=1e-02 Nkeep=256	(↑↑)	C3	2.49e-01	–	report
V3c	T=1e-02 Nkeep=256	(↑↑)	C4	5.78e-01	–	report
V3c	T=1e-02 Nkeep=256	(↑↑)	C4 (BigFloat, default chain)	5.78e-01	–	report
V3c	T=1e-02 Nkeep=256	(↑↑)	C5	0.00e+00	–	report
V3c	T=1e-02 Nkeep=256	(↑↑)	C6	1.58e-15	–	report
V3c	T=1e-02 Nkeep=256	(↑↑)	C7	2.87e-08	–	report
V3c	T=1e-02 Nkeep=256	(↑↑)	C8	1.49e-04	–	report
V3c	T=1e-02 Nkeep=256	(↑↑)	K̃ vs Nkeep=512 (rel.)	6.76e-03	–	report
V3c	T=1e-02 Nkeep=512	(↑↓)	C1	2.19e-06	–	report
V3c	T=1e-02 Nkeep=512	(↑↓)	C2	5.81e-13	–	report
V3c	T=1e-02 Nkeep=512	(↑↓)	C2 blocks n_agree/L	6.73e-02	–	report
V3c	T=1e-02 Nkeep=512	(↑↓)	C3	2.82e-01	–	report
V3c	T=1e-02 Nkeep=512	(↑↓)	C4	5.38e-01	–	report
V3c	T=1e-02 Nkeep=512	(↑↓)	C4 (BigFloat, default chain)	5.38e-01	–	report
V3c	T=1e-02 Nkeep=512	(↑↓)	C5	0.00e+00	–	report
V3c	T=1e-02 Nkeep=512	(↑↓)	C6	2.11e-15	–	report
V3c	T=1e-02 Nkeep=512	(↑↓)	C7	4.67e-08	–	report
V3c	T=1e-02 Nkeep=512	(↑↓)	C8	8.88e-05	–	report
V3c	T=1e-02 Nkeep=512	(↑↓)	K̃ vs Nkeep=512 (rel.)	0.00e+00	–	report
V3c	T=1e-02 Nkeep=512	(↑↑)	C1	2.19e-06	–	report
V3c	T=1e-02 Nkeep=512	(↑↑)	C2	5.81e-13	–	report
V3c	T=1e-02 Nkeep=512	(↑↑)	C2 blocks n_agree/L	6.73e-02	–	report
V3c	T=1e-02 Nkeep=512	(↑↑)	C3	2.82e-01	–	report
V3c	T=1e-02 Nkeep=512	(↑↑)	C4	5.38e-01	–	report
V3c	T=1e-02 Nkeep=512	(↑↑)	C4 (BigFloat, default chain)	5.38e-01	–	report
V3c	T=1e-02 Nkeep=512	(↑↑)	C5	0.00e+00	–	report
V3c	T=1e-02 Nkeep=512	(↑↑)	C6	1.21e-15	–	report
V3c	T=1e-02 Nkeep=512	(↑↑)	C7	2.43e-08	–	report
V3c	T=1e-02 Nkeep=512	(↑↑)	C8	1.24e-04	–	report
V3c	T=1e-02 Nkeep=512	(↑↑)	K̃ vs Nkeep=512 (rel.)	0.00e+00	–	report
V3c	T=1e-04 Nkeep=64	(↑↓)	C1	1.37e-02	–	report
V3c	T=1e-04 Nkeep=64	(↑↓)	C2	1.26e-13	–	report
V3c	T=1e-04 Nkeep=64	(↑↓)	C2 blocks n_agree/L	6.68e-02	–	report
V3c	T=1e-04 Nkeep=64	(↑↓)	C3	5.98e-01	–	report
V3c	T=1e-04 Nkeep=64	(↑↓)	C4	5.01e-01	–	report
V3c	T=1e-04 Nkeep=64	(↑↓)	C4 (BigFloat, default chain)	5.01e-01	–	report
V3c	T=1e-04 Nkeep=64	(↑↓)	C5	0.00e+00	–	report
V3c	T=1e-04 Nkeep=64	(↑↓)	C6	1.39e-05	–	report
V3c	T=1e-04 Nkeep=64	(↑↓)	C7	1.46e-02	–	report
V3c	T=1e-04 Nkeep=64	(↑↓)	C8	3.69e-01	–	report
V3c	T=1e-04 Nkeep=64	(↑↓)	K̃ vs Nkeep=512 (rel.)	3.67e-01	–	report
V3c	T=1e-04 Nkeep=64	(↑↑)	C1	1.37e-02	–	report
V3c	T=1e-04 Nkeep=64	(↑↑)	C2	1.26e-13	–	report
V3c	T=1e-04 Nkeep=64	(↑↑)	C2 blocks n_agree/L	6.68e-02	–	report
V3c	T=1e-04 Nkeep=64	(↑↑)	C3	5.98e-01	–	report
V3c	T=1e-04 Nkeep=64	(↑↑)	C4	5.01e-01	–	report
V3c	T=1e-04 Nkeep=64	(↑↑)	C4 (BigFloat, default chain)	5.01e-01	–	report
V3c	T=1e-04 Nkeep=64	(↑↑)	C5	0.00e+00	–	report
V3c	T=1e-04 Nkeep=64	(↑↑)	C6	1.39e-05	–	report
V3c	T=1e-04 Nkeep=64	(↑↑)	C7	2.25e-02	–	report
V3c	T=1e-04 Nkeep=64	(↑↑)	C8	5.79e-01	–	report
V3c	T=1e-04 Nkeep=64	(↑↑)	K̃ vs Nkeep=512 (rel.)	4.31e-01	–	report
V3c	T=1e-04 Nkeep=128	(↑↓)	C1	4.40e-04	–	report
V3c	T=1e-04 Nkeep=128	(↑↓)	C2	2.17e-14	–	report
V3c	T=1e-04 Nkeep=128	(↑↓)	C2 blocks n_agree/L	6.05e-02	–	report
V3c	T=1e-04 Nkeep=128	(↑↓)	C3	4.20e-01	–	report

```
V3c   T=1e-04 Nkeep=128         (↑↓)    C4                              5.69e-01        –  report
V3c   T=1e-04 Nkeep=128         (↑↓)    C4 (BigFloat, default chain)    5.69e-01        –  report
V3c   T=1e-04 Nkeep=128         (↑↓)    C5                              0.00e+00        –  report
V3c   T=1e-04 Nkeep=128         (↑↓)    C6                              6.37e-07        –  report
V3c   T=1e-04 Nkeep=128         (↑↓)    C7                              2.41e-03        –  report
V3c   T=1e-04 Nkeep=128         (↑↓)    C8                              3.15e-03        –  report
V3c   T=1e-04 Nkeep=128         (↑↓)    K̃ vs Nkeep=512 (rel.)           5.27e-03        –  report
V3c   T=1e-04 Nkeep=128         (↑↑)    C1                              4.40e-04        –  report
V3c   T=1e-04 Nkeep=128         (↑↑)    C2                              2.17e-14        –  report
V3c   T=1e-04 Nkeep=128         (↑↑)    C2 blocks n_agree/L             6.05e-02        –  report
V3c   T=1e-04 Nkeep=128         (↑↑)    C3                              4.20e-01        –  report
V3c   T=1e-04 Nkeep=128         (↑↑)    C4                              5.69e-01        –  report
V3c   T=1e-04 Nkeep=128         (↑↑)    C4 (BigFloat, default chain)    5.69e-01        –  report
V3c   T=1e-04 Nkeep=128         (↑↑)    C5                              0.00e+00        –  report
V3c   T=1e-04 Nkeep=128         (↑↑)    C6                              6.37e-07        –  report
V3c   T=1e-04 Nkeep=128         (↑↑)    C7                              2.41e-03        –  report
V3c   T=1e-04 Nkeep=128         (↑↑)    C8                              2.85e-03        –  report
V3c   T=1e-04 Nkeep=128         (↑↑)    K̃ vs Nkeep=512 (rel.)           5.92e-03        –  report
V3c   T=1e-04 Nkeep=256         (↑↓)    C1                              3.37e-05        –  report
V3c   T=1e-04 Nkeep=256         (↑↓)    C2                              3.43e-14        –  report
V3c   T=1e-04 Nkeep=256         (↑↓)    C2 blocks n_agree/L             6.04e-02        –  report
V3c   T=1e-04 Nkeep=256         (↑↓)    C3                              2.60e-01        –  report
V3c   T=1e-04 Nkeep=256         (↑↓)    C4                              6.43e-01        –  report
V3c   T=1e-04 Nkeep=256         (↑↓)    C4 (BigFloat, default chain)    6.43e-01        –  report
V3c   T=1e-04 Nkeep=256         (↑↓)    C5                              0.00e+00        –  report
V3c   T=1e-04 Nkeep=256         (↑↓)    C6                              2.59e-06        –  report
V3c   T=1e-04 Nkeep=256         (↑↓)    C7                              2.53e-04        –  report
V3c   T=1e-04 Nkeep=256         (↑↓)    C8                              7.30e-04        –  report
V3c   T=1e-04 Nkeep=256         (↑↓)    K̃ vs Nkeep=512 (rel.)           2.30e-03        –  report
V3c   T=1e-04 Nkeep=256         (↑↑)    C1                              3.37e-05        –  report
V3c   T=1e-04 Nkeep=256         (↑↑)    C2                              3.43e-14        –  report
V3c   T=1e-04 Nkeep=256         (↑↑)    C2 blocks n_agree/L             6.04e-02        –  report
V3c   T=1e-04 Nkeep=256         (↑↑)    C3                              2.60e-01        –  report
V3c   T=1e-04 Nkeep=256         (↑↑)    C4                              6.43e-01        –  report
V3c   T=1e-04 Nkeep=256         (↑↑)    C4 (BigFloat, default chain)    6.43e-01        –  report
V3c   T=1e-04 Nkeep=256         (↑↑)    C5                              0.00e+00        –  report
V3c   T=1e-04 Nkeep=256         (↑↑)    C6                              2.59e-06        –  report
V3c   T=1e-04 Nkeep=256         (↑↑)    C7                              2.31e-04        –  report
V3c   T=1e-04 Nkeep=256         (↑↑)    C8                              5.67e-04        –  report
V3c   T=1e-04 Nkeep=256         (↑↑)    K̃ vs Nkeep=512 (rel.)           2.09e-03        –  report
V3c   T=1e-04 Nkeep=512         (↑↓)    C1                              1.06e-04        –  report
V3c   T=1e-04 Nkeep=512         (↑↓)    C2                              9.21e-14        –  report
V3c   T=1e-04 Nkeep=512         (↑↓)    C2 blocks n_agree/L             5.70e-02        –  report
V3c   T=1e-04 Nkeep=512         (↑↓)    C3                              2.82e-01        –  report
V3c   T=1e-04 Nkeep=512         (↑↓)    C4                              4.58e-01        –  report
V3c   T=1e-04 Nkeep=512         (↑↓)    C4 (BigFloat, default chain)    4.58e-01        –  report
V3c   T=1e-04 Nkeep=512         (↑↓)    C5                              0.00e+00        –  report
V3c   T=1e-04 Nkeep=512         (↑↓)    C6                              2.75e-06        –  report
V3c   T=1e-04 Nkeep=512         (↑↓)    C7                              2.27e-04        –  report
V3c   T=1e-04 Nkeep=512         (↑↓)    C8                              6.95e-04        –  report
V3c   T=1e-04 Nkeep=512         (↑↓)    K̃ vs Nkeep=512 (rel.)           0.00e+00        –  report
V3c   T=1e-04 Nkeep=512         (↑↑)    C1                              1.06e-04        –  report
V3c   T=1e-04 Nkeep=512         (↑↑)    C2                              9.21e-14        –  report
V3c   T=1e-04 Nkeep=512         (↑↑)    C2 blocks n_agree/L             5.70e-02        –  report
V3c   T=1e-04 Nkeep=512         (↑↑)    C3                              2.82e-01        –  report
V3c   T=1e-04 Nkeep=512         (↑↑)    C4                              4.58e-01        –  report
V3c   T=1e-04 Nkeep=512         (↑↑)    C4 (BigFloat, default chain)    4.58e-01        –  report
V3c   T=1e-04 Nkeep=512         (↑↑)    C5                              0.00e+00        –  report
V3c   T=1e-04 Nkeep=512         (↑↑)    C6                              2.75e-06        –  report
V3c   T=1e-04 Nkeep=512         (↑↑)    C7                              2.03e-04        –  report
V3c   T=1e-04 Nkeep=512         (↑↑)    C8                              4.49e-04        –  report
V3c   T=1e-04 Nkeep=512         (↑↑)    K̃ vs Nkeep=512 (rel.)           0.00e+00        –  report

532 rows: 252 pass, 0 FAIL, 280 report
```

Failures and open findings

Generated list of all FAIL rows, followed by the findings of Sec. 7.

In [4]:

```
if isempty(FAILS)
    println("Failures: none")
else
```

```
    println("Failures (guaranteed check beyond tolerance):")
    for r in FAILS
        @printf("  %-5s %-26s %-7s %-18s %.2e > %.0e\n", r.phase, r.case, r.chan, r.check, r.value, r.tol)
    end
    fc = unique([r.check for r in FAILS])
    println("checks affected: ", join(fc, ", "))
end
println("\nOpen findings (Sec. 7): F1 (the unstabilized chain leaks, C4 report under A1); F2 (C2 Jacobi blocks agree only on",
        " leading blocks, report under A3); F3 (K ≡ 0 in the atom's (↑↑) channel); V3c (Sec. 6, report only).")
```

```
Failures: none

Open findings (Sec. 7): F1 (the unstabilized chain leaks, C4 report under A1); F2 (C2 Jacobi blocks agree only on leading blocks, report under A3); F3 (K ≡ 0 in the atom's (↑↑) channel); V3c (Sec. 6, report only).
```

1. Conventions

Checked before any comparison is trusted (task sheet Secs. 2.2, 2.3, 2.5, 5.1):

1. PSF convention (Kugler Eq. (28)) on a 2p example by explicit Lehmann summation: expected agreement to rounding.
2. ζ signs of $(p_1, \ldots, p_6)$ for $(q_{12}, T^{(3)}, T^{(4)})$ from `HA.permutation_sign(p, [false,true,true])`: expected $(+,-,+,-,-,+)$.
3. Odd bin map, $c(-E) = -c(E)$, for the ExactGrid and the LogGrid.
4. The (I3) index map with the barred indices, with one worked entry.
5. The frequency convention of `regular_sum`: `z = [iω12, iω3, iω4]` must equal the [3p] Sec. III.3 form $\sum_p \zeta^p F_p[\vec{\mathcal{O}}_p](i\alpha_{\bar{1}}, -i\alpha_{\bar{3}})$, evaluated independently with the Liouvillian superoperator in the eigenbasis (`HA.liouvillian_regular`).

In [5]:
```
m = HA.HubbardAtomModel(U_AT, 10.0); sp = HA.atom_spectrum(m; εd = ED_AT)
ops = HA.vk_atom_operators(m; spins = [:up], i12 = (:dn, :dn))
@printf("1. PSF convention: max |Δw| = %.1e (S[d,d†]), %.1e (S[q,d†]), %.1e (S[B†,B])\n",
        HA.psf_convention_check(sp, ops.Xd[1], ops.X[1]), HA.psf_convention_check(sp, ops.Xd[2], ops.X[1]),
        HA.psf_convention_check(sp, ops.Bqd, ops.Bq))
println("2. ζ(p1..p6) = ", HA.vk_zeta(), "   expected [1, -1, 1, -1, -1, 1]: ", HA.vk_zeta() == [1, -1, 1, -1, -1, 1])
ge = R["V2 Nch=1 B=0.0 updn exact"][:data].meta[:grid]
bg, _ = vk_loggrid(vk_grid())
xs = vcat(10.0 .^ range(-7, 1.5; length = 400), [0.0, 1e-15, 3e-14])
println("3. odd bin map: LogGrid ", all(x -> HA.bin_of(bg, -x) == bg.bar[HA.bin_of(bg, x)], xs),
        "; ExactGrid (V2a Nch=1, ", length(ge.b), " bins) ", all(β -> HA.bin_of(ge, -ge.b[β]) == ge.bar[β], eachindex(ge.b)),
        "; within-cluster spread ", @sprintf("%.1e", ge.spread))
```

```
1. PSF convention: max |Δw| = 0.0e+00 (S[d,d†]), 0.0e+00 (S[q,d†]), 0.0e+00 (S[B†,B])
2. ζ(p1..p6) =
[1, -1, 1, -1, -1, 1]   expected [1, -1, 1, -1, -1, 1]: true
3. odd bin map: LogGrid
true; ExactGrid (V2a Nch=1, 625 bins) true; within-cluster spread 2.6e-15
```

In [6]:
```
# 4. (I3) worked entry: s2[α,β][a,b] = S(ᾱ, β̄)[X_a, B, X_b†] for the atom, channel (↑↑) (B = q↑↑), T = 0.1
#    (in the (↑↓) channel of the atom the family s2 vanishes identically)
d = R["V1 atom T=0.1 upup"][:data]
ops = HA.vk_atom_operators(m; spins = [:up], i12 = (:up, :up))
println("(I3): s1 = S(α,β)[X_a†,B,X_b], s2 = S(ᾱ,β̄)[X_a,B,X_b†], s3 = S(α,β)[B,x,X_b], s4 = S(ᾱ,β)[X_a,x,B],")
println("      s5 = S(α,β̄)[B,y,X_b†], s6 = S(α,β)[X_a†,y,B]")
(k, blk) = first(sort(collect(d.s2); by = first))
```

```
a, b = Tuple(argmax(abs.(blk)))
α, β = k
names = ["d↑", "q↑"]
@printf("worked entry: s2[(α,β) = (%d,%d)][a,b] = [%s,%s] = %.6f, i.e. the PSF S[%s, B, %s] at (ε1,ε2) = (b_α̃, b_β̃) = (%.4f, %.4f)\n",
        α, β, names[a], names[b], real(blk[a, b]), names[a], names[b] == "d↑" ? "d" : "q", d.b[d.bar[α]], d.b[d.bar[β]])
ts = HA.psf(sp, (ops.X[a], ops.Bq, ops.Xd[b]))
w = sum(t.weight for t in ts if abs(t.position[1] - d.b[d.bar[α]]) < 1e-10 && abs(t.position[2] - d.b[d.bar[β]]) < 1e-10; init = 0.0im)
@printf("direct PSF sum at that position: %.6f  (difference %.1e)\n", real(w), abs(w - blk[a, b]))
```

```
(I3): s1 = S(α,β)[X_a↑,B,X_b], s2 = S(α̃,β̃)[X_a,B,X_b↑], s3 = S(α,β)[B,x,X_b], s4 = S(α̃,β)[X_a,x,B],
     s5 = S(α,β̃)[B,y,X_b↑], s6 = S(α,β)[X_a↑,y,B]
worked entry: s2[(α,β) = (5,5)][a,b] = [d↑,d↑] = 0.000445, i.e. the PSF S[d↑, B, d] at (ε1,ε2) = (b_α̃, b_β̃) = (-0.7000, -0.7000)
direct PSF sum at that position: 0.000445  (difference 0.0e+00)
```

In [7]:

```
# 5. regular_sum frequency convention vs [3p] Sec. III.3 (Liouvillian form)
err = 0.0
for (n3, n4) in ((0, 0), (2, -5), (-3, 1), (4, 4)), (x, y) in ((ops.q[1], ops.X[2]), (ops.d[1], ops.X[1]), (ops.q[1], ops.X[1]))
    ω3 = HA.vk_fermionic(n3, 0.1); ω4 = HA.vk_fermionic(n4, 0.1); ω12 = -ω3 - ω4
    rs = HA.regular_sum(sp, (ops.Bq, x, y), ComplexF64[im * ω12, im * ω3, im * ω4]; is_fermionic = [false, true, true])
    global err = max(err, abs(rs - HA.liouvillian_regular(sp, (ops.Bq, x, y), [ω12, ω3, ω4])))
end
@printf("5. regular_sum(z = [iω12, iω3, iω4]) vs Σ_p ζ^p F_p(iα_p̃1, -iα_p̃3): max |Δ| = %.1e over 12 points\n", err)
```

```
5. regular_sum(z = [iω12, iω3, iω4]) vs Σ_p ζ^p F_p(iα_p̃1, -iα_p̃3): max |Δ| = 6.9e-17 over 12 points
```

2. Hubbard atom (V1)

$U = 1$, $\varepsilon_d = -0.3$, $T \in \{0.1, 0.5\}$, HubbardAtom ED on the ExactGrid. Expected: the closed-form one-level chain of [VD] App. F, $\bar{\mathbb{J}} = \varepsilon_\sigma + U(1-n)$, $\bar{N}^2 = U^2 n(1-n)$, $\sigma = Un$, $n = \langle n_{\bar\sigma} \rangle$, chain length 1; all checks to the V1 tolerances. The bosonic legs of the atom are trivial (B is a conserved density, $\mathbb{J}_B$ is 1×1 with entry 0). In the (↑↑) channel $B = Un_\downarrow$ commutes with H, $d_\uparrow$, $q_\uparrow$, so $K^{(12)} \equiv 0$ at $\omega_{12} \neq 0$ while the six permutation terms are $O(1)$; C8 is then measured on the scale of the permutation terms (also shown: the per-permutation agreement, which holds term by term).

In [8]:

```
@printf("%-26s %3s %10s %10s %10s %10s %9s %9s %9s %9s %9s %9s %9s %9s %9s\n", "case", "L", "J̄", "J̄ closed", "N̄²", "N̄² closed",
        "C1", "C2", "C3", "C4", "C5", "C6", "C7", "C8")
for T in (0.1, 0.5), chan in (:updn, :upup)
    o = R["V1 atom T=$T $chan"]; c = o[:checks]; cf = o[:closed]; ch = o[:res].ch
    @printf("%-26s %3d %10.6f %10.6f %10.6f %10.6f %9.1e %9.1e %9.1e %9.1e %9.1e %9.1e %9.1e %9.1e %9.1e%s\n", "T=$T $(chs(chan))",
            cf.L, real(ch.Jbar.eps[1][1]), cf.J, real(ch.Nbar[1])^2, cf.N2, c[:C1], c[:C2], c[:C3], c[:C4], c[:C5], c[:C6],
            c[:C7], c[:C8], c[:C8_parts].degenerate ? " †" : "")
end
println("\nC8 detail (abs = max|K̃ - K^direct|, perm = per-permutation error / max|K_p|):")
for T in (0.1, 0.5), chan in (:updn, :upup)
    p = R["V1 atom T=$T $chan"][:checks][:C8_parts]
    @printf("  T=%-4s %s: max|K^direct| = %.2e, max|K_p| = %.2e, abs = %.1e, rel = %.1e, rel_perm = %.1e, perm = %.1e\n",
            T, chs(chan), p.maxKd, p.maxKp, p.abs, p.rel, p.rel_perm, p.perm)
end
println("\nBigFloat (128 bit) particle chain vs Float64: ",
        join([@sprintf("T=%s %s: |ΔJ̄| = %.1e", T, chs(ch), maximum(R["V1 atom T=$T $ch"][:big].dev)) for T in (0.1, 0.5), ch in (:updn, :upup)], "; "))
```

```
case                       L          J̄   J̄ closed         N̄²  N̄² closed        C1        C2        C3        C4        C5        C6        C7        C8
T=0.1 (↑↓)                 1   0.211917   0.211917   0.249858   0.249858   8.0e-16   2.8e-14   1.7e-15   2.2e-16   0.0e+00   1.0e-16   3.4e-16   3.8e-16
T=0.1 (↑↑)                 1   0.211917   0.211917   0.249858   0.249858   8.0e-16   2.8e-14   1.7e-15   2.2e-16   0.0e+00   1.0e-16   3.7e-16   3.2e-16 †
T=0.5 (↑↓)                 1   0.254056   0.254056   0.247078   0.247078   5.5e-16   5.4e-15   5.7e-15   1.1e-16   0.0e+00   1.1e-16   1.6e-16   1.5e-16
T=0.5 (↑↑)                 1   0.254056   0.254056   0.247078   0.247078   5.5e-16   5.4e-15   5.7e-15   1.1e-16   0.0e+00   1.1e-16   1.6e-16   1.5e-16 †

C8 detail (abs = max|K̃ - K^direct|, perm = per-permutation error / max|K_p|):
  T=0.1  (↑↓): max|K^direct| = 1.74e+00, max|K_p| = 9.30e-01, abs = 6.7e-16, rel = 3.8e-16, rel_perm = 7.2e-16, perm = 3.7e-16
```

```
  T=0.1  (↑↑): max|K^direct| = 1.79e-16, max|K_p| = 1.09e+00, abs = 3.4e-16, rel = 1.9e+00, rel_perm = 3.2e-16, perm = 2.9e-16
  T=0.5  (↑↓): max|K^direct| = 9.76e-02, max|K_p| = 5.88e-02, abs = 1.4e-17, rel = 1.5e-16, rel_perm = 2.4e-16, perm = 1.2e-16
  T=0.5  (↑↑): max|K^direct| = 3.74e-18, max|K_p| = 4.34e-02, abs = 6.3e-18, rel = 1.7e+00, rel_perm = 1.5e-16, perm = 2.5e-16

BigFloat (128 bit) particle chain vs Float64:
T=0.1 (↑↓): |ΔJ̃| = 8.2e-17; T=0.5 (↑↓): |ΔJ̃| = 2.7e-17; T=0.1 (↑↑): |ΔJ̃| = 8.2e-17; T=0.5 (↑↑): |ΔJ̃| = 2.7e-17
```

In [9]:

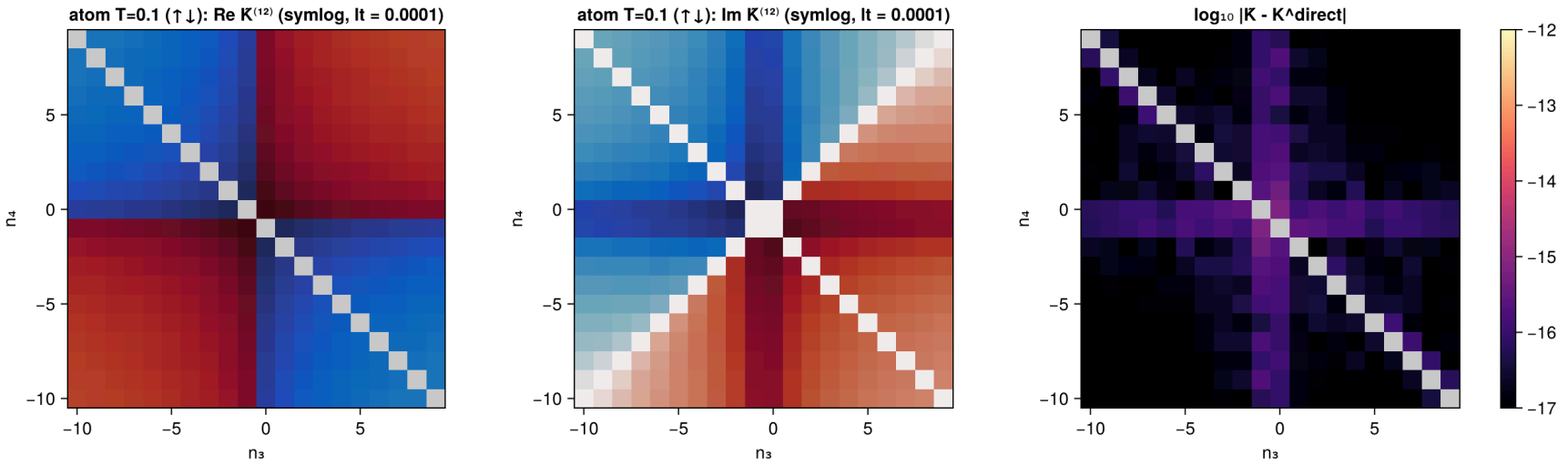
```
# N = 2 block test: spin-unresolved atom, X = (d†↑, d†↓, q†↑, q†↓), B = q↓↓, vs the N = 1 runs
for h in (0.0, 0.07)
    o2 = R["V1 atom N=2 h=$h"]; ou = R["V1 atom N=1 up h=$h"]; od = R["V1 atom N=1 dn h=$h"]
    b2 = o2[:checks][:box]; bu = ou[:checks][:box]; bd = od[:checks][:box]
    m_ = .!isnan.(real.(bu.Kd))
    zs = HA.vk_testpoints(o2[:data])
    dΣ = maximum(z -> maximum(abs, HA.sigma_chain(o2[:res].ch, z) - Diagonal([HA.sigma_chain(ou[:res].ch, z)[1], HA.sigma_chain(od[:res].ch, z)[1]])), zs)
    @printf("h = %.2f: max|K̃↑↑(N=2) - K̃(N=1,↑)| = %.1e, max|K̃↓↓(N=2) - K̃(N=1,↓)| = %.1e, max|K̃ off-diag| = %.1e, max|ΔΣ^S| = %.1e; C1..C8: %s\n",
            h, maximum(abs, b2.Kt[1, 1, :, :][m_] - bu.Kt[m_]), maximum(abs, b2.Kt[2, 2, :, :][m_] - bd.Kt[m_]),
            max(maximum(abs, b2.Kt[1, 2, :, :][m_]), maximum(abs, b2.Kt[2, 1, :, :][m_])), dΣ,
            join([@sprintf("%.0e", o2[:checks][k]) for k in (:C1, :C2, :C3, :C4, :C5, :C6, :C7, :C8)], " "))
end
o = R["V1 atom N=2 rotated"]; bx = o[:checks][:box]; m_ = .!isnan.(real.(bx.Kd[1, 1, :, :]))
@printf("rotated complex orbitals (h = 0.07): max|K12 - K21| = %.2f (asymmetric, exercises the p2 transposition); C8 = %.1e, C1 = %.1e, C7 = %.1e\n",
        maximum(abs, bx.Kd[1, 2, :, :][m_] - bx.Kd[2, 1, :, :][m_]), o[:checks][:C8], o[:checks][:C1], o[:checks][:C7])
```

```
h = 0.00: max|K̃↑↑(N=2) - K̃(N=1,↑)| = 4.0e-16, max|K̃↓↓(N=2) - K̃(N=1,↓)| = 1.7e-16, max|K̃ off-diag| = 0.0e+00, max|ΔΣ^S| = 4.5e-15; C1..C8: 9e-16 2e-14 2e-15 0e+00 0e+00 1e-16 7e-16 2e-16
h = 0.07: max|K̃↑↑(N=2) - K̃(N=1,↑)| = 4.5e-16, max|K̃↓↓(N=2) - K̃(N=1,↓)| = 7.1e-16, max|K̃ off-diag| = 0.0e+00, max|ΔΣ^S| = 1.7e-15; C1..C8: 2e-15 3e-14 2e-15 3e-16 0e+00 2e-16 5e-16 8e-16
rotated complex orbitals (h = 0.07): max|K12 - K21| = 0.36 (asymmetric, exercises the p2 transposition); C8 = 4.6e-16, C1 = 1.1e-15, C7 = 8.0e-16
```

In [10]:

```
function symlog_heat!(ax, n3s, n4s, M; lt = 1e-4)
    heatmap!(ax, n3s, n4s, map(x -> isnan(x) ? NaN : HA.symlog(x, lt), M); colormap = :balance, colorrange = (-1, 1), nan_color = :gray80)
end
function K_heatmaps(box; title = "", lt = 1e-4)
    fig = Figure(size = (1300, 400))
    for (j, (M, ttl)) in enumerate(((real.(box.Kt), "Re K̃⁽¹²⁾"), (imag.(box.Kt), "Im K̃⁽¹²⁾")))
        ax = Axis(fig[1, j]; xlabel = "n₃", ylabel = "n₄", title = "$title: $ttl (symlog, lt = $lt)", aspect = 1)
        symlog_heat!(ax, box.n3s, box.n4s, M; lt = lt)
    end
    E = abs.(box.Kt - box.Kd)
    ax = Axis(fig[1, 3]; xlabel = "n₃", ylabel = "n₄", title = "log₁₀ |K̃ - K^direct|", aspect = 1)
    hm = heatmap!(ax, box.n3s, box.n4s, log10.(fl.(E)); colormap = :magma, colorrange = (-17, -12), nan_color = :gray80)
    Colorbar(fig[1, 4], hm)
    fig
end
savefig("atom_K_updn", K_heatmaps(R["V1 atom T=0.1 updn"][:checks][:box]; title = "atom T=0.1 (↑↓)"))
```

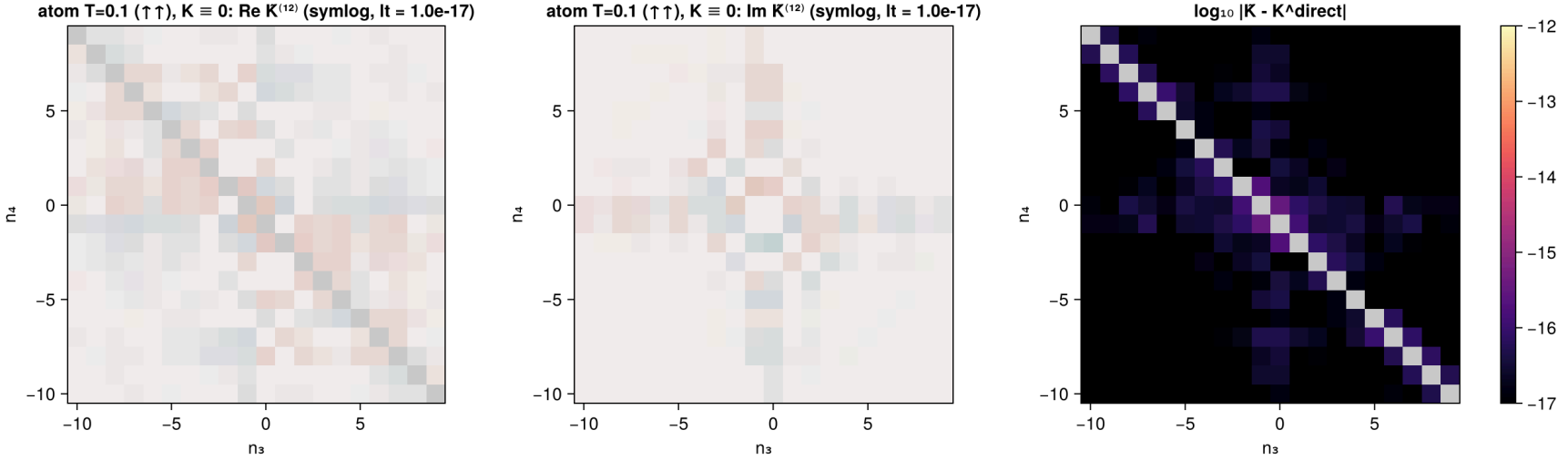
Out[10]:



In [11]:

```
savefig("atom_K_upup", K_heatmaps(R["V1 atom T=0.1 upup"][:checks][:box]; title = "atom T=0.1 (↑↑), K ≡ 0", lt = 1e-17))
```

Out[11]:



3. Wilson-chain ED (V2a, V2b)

`siam(U = 0.5, εd = -0.1, Δ = 0.1)`, chains with sites $-1..N_{\mathrm{ch}}$, $N_{\mathrm{ch}} \in \{1, 2\}$ (64 and 256 states), $T = 0.05$, with and without field $B = 0.05$, both channels. **V2a**: exact positions on an ExactGrid (`degtol = 1e-10`). **V2b**: the same ED binned on `LogGrid(emin = 1e-5, emax = 10, ndec = 16)`. Expected (Theorem 2): C1, C5–C8 guaranteed for both; C3 (Dyson) and C4 (bath leakage) guaranteed only for exact data (they use the equation of motion, which binned data violate by the bin width).

In [12]:
```
function chain_table(keys)
    @printf("%-34s %6s %5s %4s %4s %9s %9s %9s %9s %9s %9s %9s %9s %9s %9s\n", "case", "bins", "R", "L", "L_B", "C1", "C2", "C3", "C4",
            "C4 unst", "C5", "C6", "C7", "C8", "t [s]")
    for k in keys
        o = R[k]; c = o[:checks]
        @printf("%-34s %6d %5d %4d %4d %9.1e %9.1e %9.1e %9.1e %9s %9.1e %9.1e %9.1e %9.1e %9.1f\n", k, c[:nbins], c[:R], c[:L], c[:LB],
                c[:C1], c[:C2], c[:C3], c[:C4], haskey(o, :unstab) ? @sprintf("%.1e", o[:unstab].C4) : "–", c[:C5], c[:C6], c[:C7], c[:C8],
                sum(values(o[:timings])))
    end
end
println("V2a (ExactGrid):"); chain_table(["V2 Nch=$N B=$f $c exact" for N in (1, 2) for f in (0.0, 0.05) for c in (:updn, :upup)])
println("\nV2b (LogGrid):");  chain_table(["V2 Nch=$N B=$f $c log" for N in (1, 2) for f in (0.0, 0.05) for c in (:updn, :upup)])
println("\nsanity (Sec. 3.3), max over V2 cases: ")
for gk in (:exact, :log)
    S = [R["V2 Nch=$N B=$f $c $gk"][:checks][:sanity] for N in (1, 2) for f in (0.0, 0.05) for c in (:updn, :upup)]
    @printf("  %-5s herm %.1e, min eig/max‖W‖ %.1e, sum rule %.1e, 3p totals %.1e, C5 %.1e, Boltzmann-shortcut diff %.1e\n", gk,
            maximum(s.herm_err for s in S), minimum(s.min_eig_rel for s in S), maximum(s.sumrule_err for s in S),
            maximum(s.totals3_err for s in S), maximum(s.C5 for s in S), maximum(s.boltzmann_diff for s in S))
end
println("clamped positions on the LogGrid: ", sum(R["V2 Nch=$N B=$f $c log"][:data].meta[:nclamped] for N in (1, 2) for f in (0.0, 0.05) for c in (:updn, :upup)))
```

```
V2a (ExactGrid):
case                                bins     R    L  L_B        C1        C2        C3        C4   C4 unst        C5        C6        C7        C8     t [s]
V2 Nch=1 B=0.0 updn exact            625   207  203  217   2.9e-15   1.2e-13   4.4e-14   1.2e-14   8.4e-01   0.0e+00   7.5e-16   9.8e-16   4.7e-15      16.7
V2 Nch=1 B=0.0 upup exact            625   207  203  217   3.2e-15   1.1e-13   4.4e-14   1.2e-14   8.3e-01   0.0e+00   7.7e-16   8.1e-16   4.1e-14      12.9
V2 Nch=1 B=0.05 updn exact           871   282  279  307   2.2e-15   1.5e-13   6.5e-14   5.0e-16   4.7e-01   0.0e+00   1.2e-15   8.7e-16   3.2e-15      27.1
V2 Nch=1 B=0.05 upup exact           871   282  279  307   1.3e-15   1.5e-13   6.5e-14   5.0e-16   5.5e-01   0.0e+00   9.7e-16   8.4e-16   2.4e-14      27.1
V2 Nch=2 B=0.0 updn exact           7901  2616 2612 2797   4.4e-15   1.6e-13   1.0e-13   4.9e-15   2.5e-01   0.0e+00   2.7e-15   9.7e-16   4.8e-14     528.6
V2 Nch=2 B=0.0 upup exact           7901  2616 2612 2797   4.7e-15   1.2e-12   1.0e-13   4.9e-15   2.4e-01   0.0e+00   2.4e-15   5.0e-16   2.8e-13     537.2
V2 Nch=2 B=0.05 updn exact         12163  3840 3836 4483   6.7e-15   2.4e-13   1.6e-13   5.6e-15   2.0e-01   0.0e+00   5.4e-15   5.6e-16   8.2e-14    1507.7
V2 Nch=2 B=0.05 upup exact         12163  3840 3836 4483   6.0e-15   2.0e-13   1.6e-13   5.6e-15   2.0e-01   0.0e+00   5.6e-15   6.6e-16   7.4e-14    1542.9

V2b (LogGrid):
case                                bins     R    L  L_B        C1        C2        C3        C4   C4 unst        C5        C6        C7        C8     t [s]
V2 Nch=1 B=0.0 updn log              195    87   86   55   1.9e-15   5.7e-14   2.0e-01   3.7e-01         –   0.0e+00   2.9e-16   6.5e-16   2.8e-15       1.8
V2 Nch=1 B=0.0 upup log              195    87   86   55   1.9e-15   5.7e-14   2.0e-01   3.7e-01         –   0.0e+00   7.7e-16   6.5e-16   2.2e-14       1.0
V2 Nch=1 B=0.05 updn log             195   102  101   57   1.8e-15   5.6e-14   2.4e-01   5.7e-01         –   0.0e+00   8.9e-16   2.4e-14   2.7e-14       1.1
V2 Nch=1 B=0.05 upup log             195   102  101   57   1.8e-15   5.6e-14   2.4e-01   5.7e-01         –   0.0e+00   7.5e-16   3.4e-14   1.8e-13       1.1
V2 Nch=2 B=0.0 updn log              195   163  162   91   2.4e-15   2.6e-14   4.5e-01   3.8e-01         –   0.0e+00   3.9e-16   4.7e-16   2.0e-14       5.8
V2 Nch=2 B=0.0 upup log              195   163  162   91   2.4e-15   2.6e-14   4.5e-01   3.8e-01         –   0.0e+00   6.3e-16   6.2e-16   4.2e-14       5.9
V2 Nch=2 B=0.05 updn log             195   175  174   91   2.0e-15   3.8e-14   3.6e-01   3.4e-01         –   0.0e+00   7.2e-16   8.1e-16   2.7e-14       6.3
V2 Nch=2 B=0.05 upup log             195   175  174   91   2.0e-15   3.8e-14   3.6e-01   3.4e-01         –   0.0e+00   9.7e-16   8.0e-16   2.9e-14       6.2

sanity (Sec. 3.3), max over V2 cases:
  exact herm 3.2e-17, min eig/max‖W‖ -1.6e-17, sum rule 8.9e-16, 3p totals 1.9e-15, C5 0.0e+00, Boltzmann-shortcut diff 1.8e-15
  log   herm 3.2e-17, min eig/max‖W‖ -6.1e-23, sum rule 6.7e-16, 3p totals 7.8e-16, C5 0.0e+00, Boltzmann-shortcut diff 9.8e-02
clamped positions on the LogGrid: 0
```

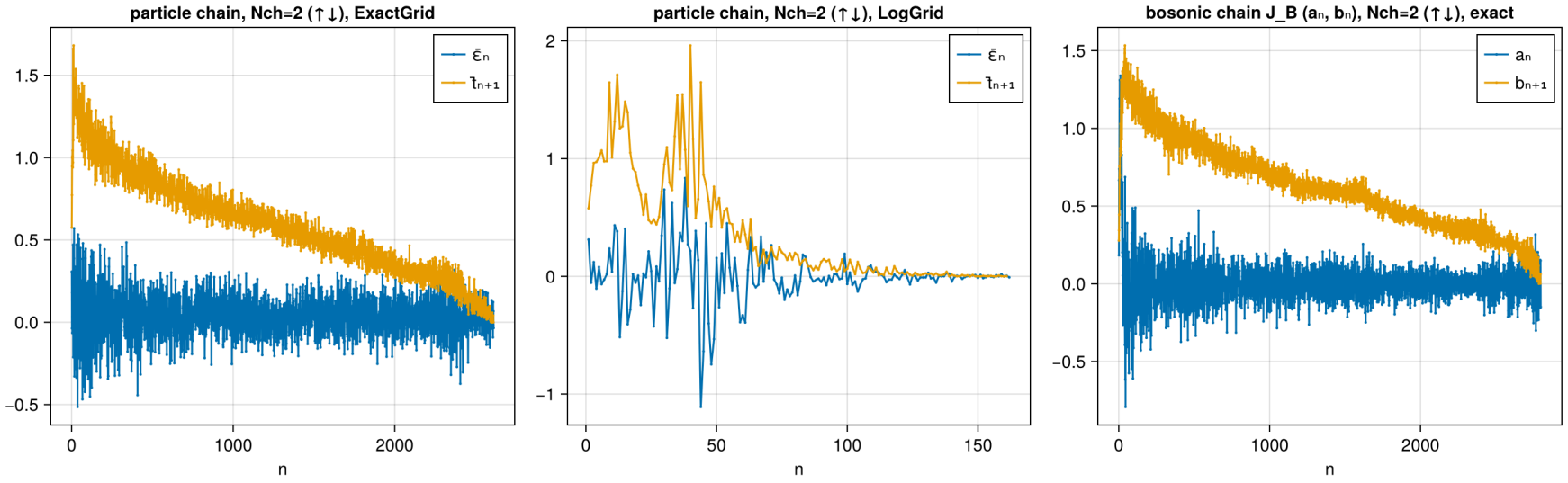
In [13]:
```
# N = 2 block test on the chain with field B = 0.05 (Nch = 1, ExactGrid)
o2 = R["V2 block N=2"]; ou = R["V2 block up"]; od = R["V2 block dn"]
b2 = o2[:checks][:box]; bu = ou[:checks][:box]; bd = od[:checks][:box]
m_ = .!isnan.(real.(bu.Kd))
zs = HA.vk_testpoints(o2[:data])
dΣ = maximum(z -> maximum(abs, HA.sigma_chain(o2[:res].ch, z) - Diagonal([HA.sigma_chain(ou[:res].ch, z)[1], HA.sigma_chain(od[:res].ch, z)[1]])), zs)
@printf("chain N=2 block (field 0.05): |ΔK̃↑↑| = %.1e, |ΔK̃↓↓| = %.1e, |K̃ off| = %.1e, |ΔΣ^S| = %.1e; L(N=2) = %d blocks, L(↑) = %d, L(↓) = %d\n",
        maximum(abs, b2.Kt[1, 1, :, :][m_] - bu.Kt[m_]), maximum(abs, b2.Kt[2, 2, :, :][m_] - bd.Kt[m_]),
        max(maximum(abs, b2.Kt[1, 2, :, :][m_]), maximum(abs, b2.Kt[2, 1, :, :][m_])), dΣ,
        o2[:checks][:L], ou[:checks][:L], od[:checks][:L])
println("spin blocks differ with field: |K̃↑↑ - K̃↓↓| = ", @sprintf("%.2e", maximum(abs, bu.Kt[m_] - bd.Kt[m_])))
```

```
chain N=2 block (field 0.05): |ΔK̄↑↑| = 1.2e-16, |ΔK̄↓↓| = 3.3e-16, |K̄ off| = 1.0e-16, |ΔΣ^S| = 2.8e-16; L(N=2) = 279 blocks, L(↑) = 279, L(↓) = 279
spin blocks differ with field: |K̄↑↑ - K̄↓↓| =
2.36e-01
```

In [14]:

```julia
# chain coefficients
fig = Figure(size = (1300, 420))
for (j, gk) in enumerate((:exact, :log))
    o = R["V2 Nch=2 B=0.0 updn $gk"]; ch = o[:res].ch
    ax = Axis(fig[1, j]; xlabel = "n", title = "particle chain, Nch=2 (↑↓), $(gk === :exact ? "ExactGrid" : "LogGrid")")
    scatterlines!(ax, 1:HA.nblocks(ch.Jbar), [real(e[1]) for e in ch.Jbar.eps]; markersize = 3, label = "ε̃ₙ")
    scatterlines!(ax, 1:(HA.nblocks(ch.Jbar) - 1), [real(t[1]) for t in ch.Jbar.lower]; markersize = 3, label = "t̃ₙ₊₁")
    axislegend(ax; position = :rt)
end
o = R["V2 Nch=2 B=0.0 updn exact"]; ch = o[:res].ch
ax = Axis(fig[1, 3]; xlabel = "n", title = "bosonic chain J_B (aₙ, bₙ), Nch=2 (↑↓), exact")
scatterlines!(ax, 1:HA.nblocks(ch.JB), [e[1] for e in ch.JB.eps]; markersize = 3, label = "aₙ")
scatterlines!(ax, 1:(HA.nblocks(ch.JB) - 1), [t[1] for t in ch.JB.lower]; markersize = 3, label = "bₙ₊₁")
axislegend(ax; position = :rt)
savefig("chain_coefficients", fig)
```

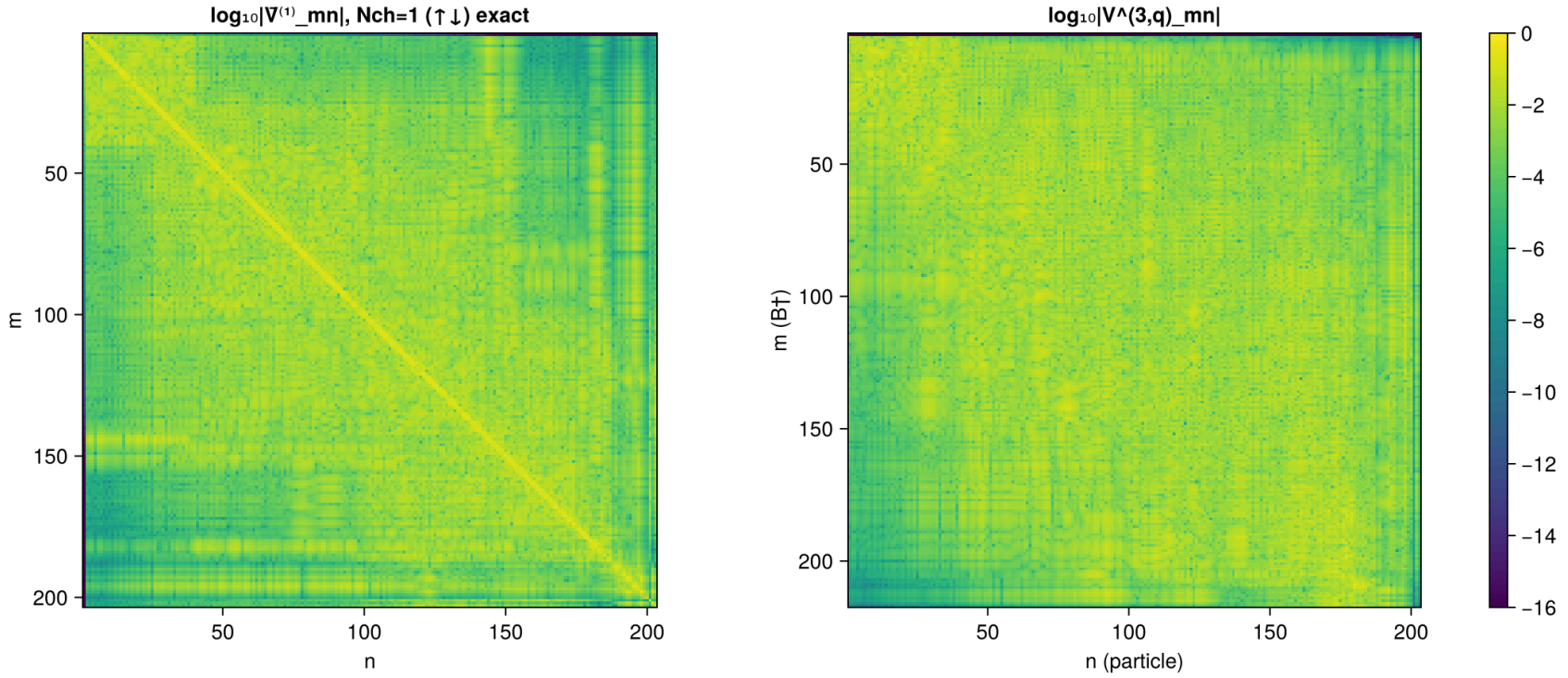
Out[14]:



In [15]:

```julia
# couplings |V~⁽¹⁾_mn| and |V^(3,q)_mn|
fig = Figure(size = (1100, 480))
o = R["V2 Nch=1 B=0.0 updn exact"]; cp = o[:res].cp
ax = Axis(fig[1, 1]; xlabel = "n", ylabel = "m", title = "log₁₀|Ṽ⁽¹⁾_mn|, Nch=1 (↑↓) exact", aspect = 1, yreversed = true)
hm = heatmap!(ax, permutedims(log10.(fl.(abs.(cp.V1)))); colormap = :viridis, colorrange = (-16, 0))
ax = Axis(fig[1, 2]; xlabel = "n (particle)", ylabel = "m (B†)", title = "log₁₀|V^(3,q)_mn|", aspect = 1, yreversed = true)
heatmap!(ax, permutedims(log10.(fl.(abs.(cp.V3[:q][1]))))); colormap = :viridis, colorrange = (-16, 0))
Colorbar(fig[1, 3], hm)
savefig("couplings", fig)
```

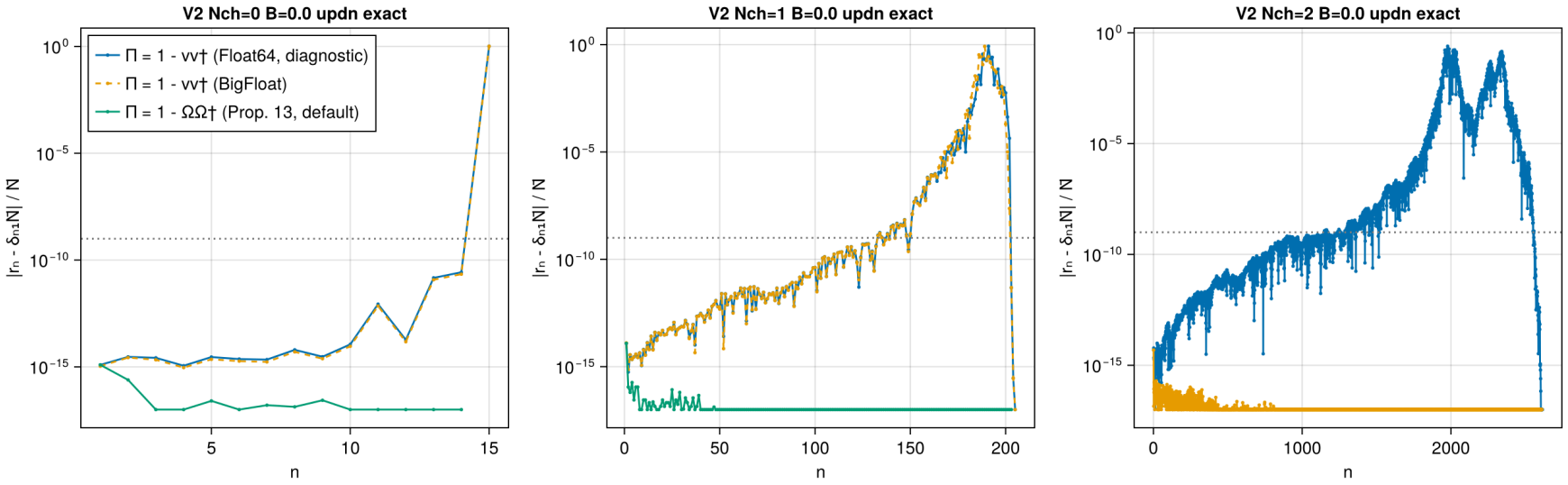
Out[15]:



In [16]:

```
# bath leakage |r_n - δ_n1 N̄| / N̄: the default (stabilized, Prop. 13) chain and the unstabilized diagnostic
fig = Figure(size = (1300, 420))
for (j, k) in enumerate(("V2 Nch=0 B=0.0 updn exact", "V2 Nch=1 B=0.0 updn exact", "V2 Nch=2 B=0.0 updn exact"))
    o = R[k]
    ax = Axis(fig[1, j]; yscale = log10, xlabel = "n", ylabel = "|rₙ - δₙ₁N̄| / N̄", title = k)
    pu = o[:unstab].profile
    scatterlines!(ax, eachindex(pu), fl.(pu); markersize = 4, label = "Π = 1 - vv† (Float64, diagnostic)")
    haskey(o, :big_unstab) && scatterlines!(ax, eachindex(o[:big_unstab].profile), fl.(o[:big_unstab].profile); markersize = 4, linestyle = :dash, label = "Π = 1 - vv† (BigFloat)")
    p = haskey(o, :chain) ? o[:chain].profile : HA.leakage_profile(o[:res].ch)
    scatterlines!(ax, eachindex(p), fl.(p); markersize = 4, label = "Π = 1 - ΩΩ† (Prop. 13, default)")
    hlines!(ax, [1e-9]; color = :gray, linestyle = :dot)
    j == 1 && axislegend(ax; position = :lt)
end
savefig("bath_leakage", fig)
```

Out[16]:



In [17]:

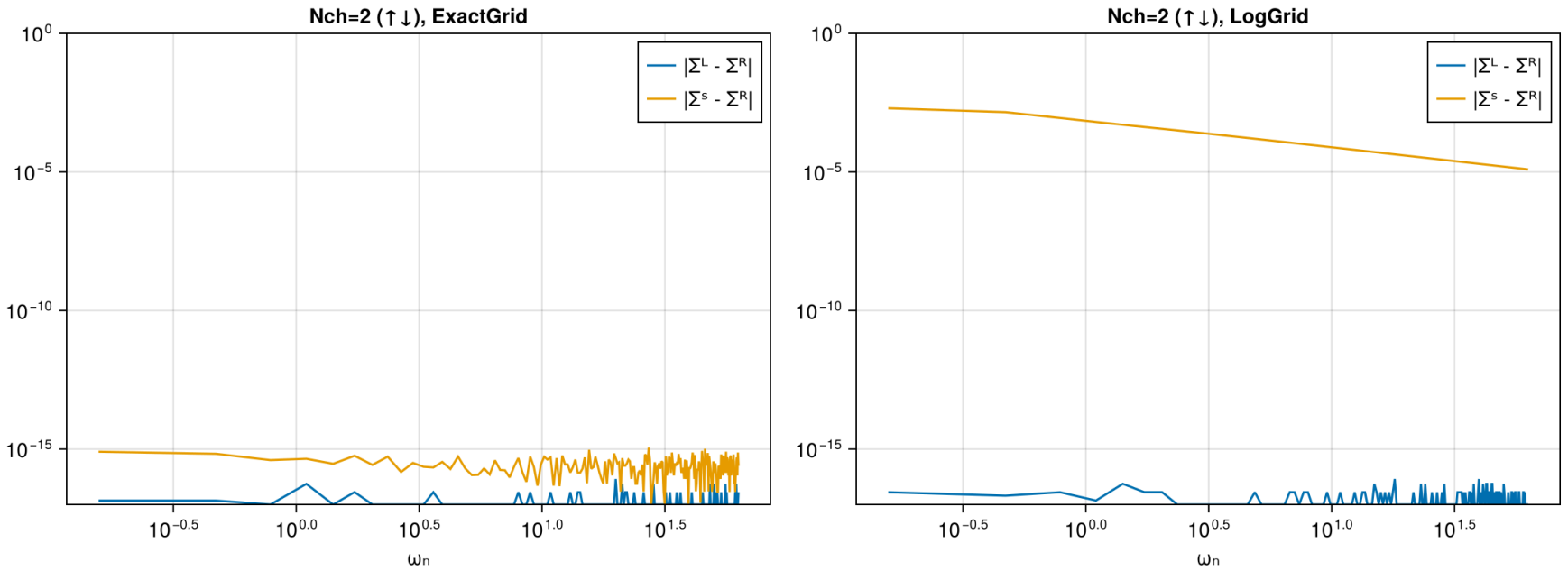
```
for k in ("V2 Nch=0 B=0.0 updn exact", "V2 Nch=1 B=0.0 updn exact", "V2 Nch=1 B=0.0 upup exact", "V2 Nch=2 B=0.0 updn exact", "V2 Nch=2 B=0.05 upup exact")
    o = R[k]; u = o[:unstab]; c = o[:checks]
    @printf("%-28s R=%5d  L=%4d (unstab %4d, BigFloat unstab %s)  C4 = %.1e (unstab %.2e, BigFloat unstab %s)  dim Ω = %d  ‖Ω†Ũ‖ = %.1e  ΔΣ^S = %.1e  ΔK̃ = %.1e\n",
            k, c[:R], c[:L], u.L, haskey(o, :big_unstab) ? string(o[:big_unstab].L) : "–", c[:C4], u.C4,
            haskey(o, :big_unstab) ? @sprintf("%.2e", o[:big_unstab].C4) : "–", c[:Omega_dim], c[:Omega_overlap], u.dSigma, u.dK)
end
```

```
V2 Nch=0 B=0.0 updn exact    R=   16  L=  14 (unstab   15, BigFloat unstab 15)  C4 = 1.2e-15 (unstab 1.03e+00, BigFloat unstab 1.03e+00)  dim Ω = 2  ‖Ω†Ũ‖ = 1.3e-15  ΔΣ^S = 3.1e-16  ΔK̃ = 1.0e-15
V2 Nch=1 B=0.0 updn exact    R=  207  L= 203 (unstab  205, BigFloat unstab 205)  C4 = 1.2e-14 (unstab 8.45e-01, BigFloat unstab 8.35e-01)  dim Ω = 3  ‖Ω†Ũ‖ = 1.3e-14  ΔΣ^S = 2.8e-16  ΔK̃ = 1.7e-15
V2 Nch=1 B=0.0 upup exact    R=  207  L= 203 (unstab  205, BigFloat unstab 205)  C4 = 1.2e-14 (unstab 8.30e-01, BigFloat unstab 8.14e-01)  dim Ω = 3  ‖Ω†Ũ‖ = 1.3e-14  ΔΣ^S = 3.5e-16  ΔK̃ = 1.4e-14
V2 Nch=2 B=0.0 updn exact    R= 2616  L=2612 (unstab 2615, BigFloat unstab –)  C4 = 4.9e-15 (unstab 2.48e-01, BigFloat unstab –)  dim Ω = 4  ‖Ω†Ũ‖ = 1.0e-14  ΔΣ^S = 3.1e-16  ΔK̃ = 1.7e-15
V2 Nch=2 B=0.05 upup exact   R= 3840  L=3836 (unstab 3839, BigFloat unstab –)  C4 = 5.6e-15 (unstab 2.01e-01, BigFloat unstab –)  dim Ω = 4  ‖Ω†Ũ‖ = 1.3e-14  ΔΣ^S = 2.5e-16  ΔK̃ = 6.0e-15
```

In [18]:

```
# |Σ^L - Σ^R| and |Σ^S - Σ^R| on the Matsubara axis, exact vs binned (why Lemma 9 is needed)
fig = Figure(size = (1100, 420))
for (j, gk) in enumerate((:exact, :log))
    o = R["V2 Nch=2 B=0.0 updn $gk"]; d = o[:data]; ch = o[:res].ch
    ns = 0:200
    LR = Float64[]; SR = Float64[]
    for n in ns
        ΣR, ΣL, ΣS = HA.sigmas(d, ch, im * HA.vk_fermionic(n, d.T))
        push!(LR, abs(ΣL[1] - ΣR[1])); push!(SR, abs(ΣS[1] - ΣR[1]))
    end
    ax = Axis(fig[1, j]; yscale = log10, xscale = log10, xlabel = "ωₙ", title = "Nch=2 (↑↓), $(gk === :exact ? "ExactGrid" : "LogGrid")")
    ωs = [HA.vk_fermionic(n, d.T) for n in ns]
    lines!(ax, ωs, fl.(LR); label = "|Σᴸ - Σᴿ|")
    lines!(ax, ωs, fl.(SR); label = "|Σˢ - Σᴿ|")
    ylims!(ax, 1e-17, 1)
    axislegend(ax; position = :rt)
end
savefig("sigma_estimators", fig)
```

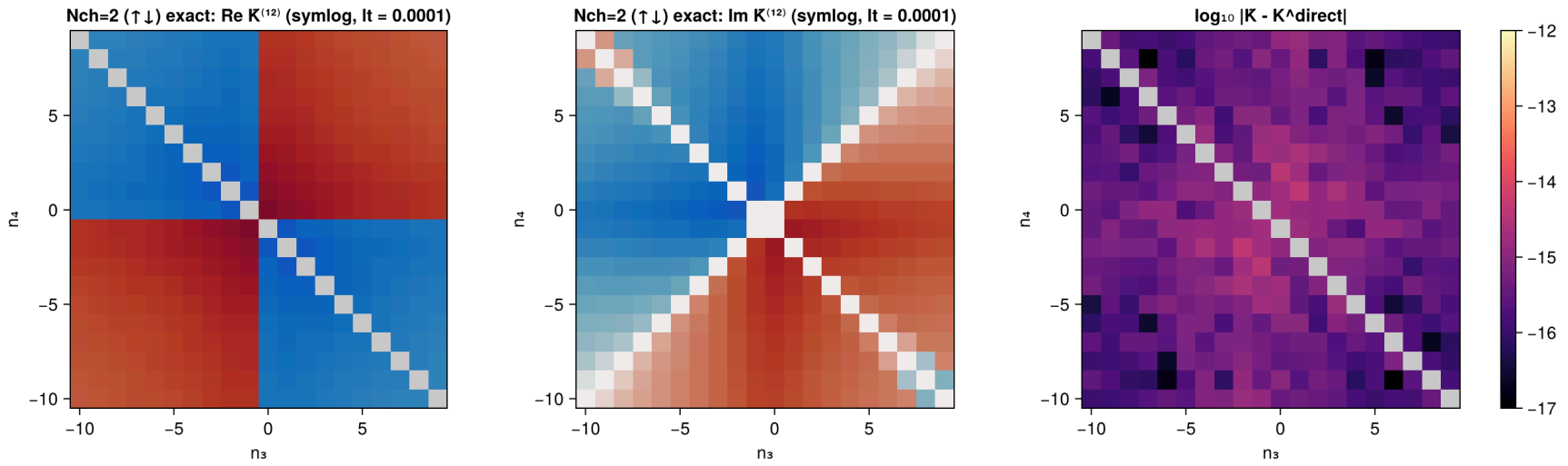
Out[18]:



In [19]:

```
savefig("chain_K_exact", K_heatmaps(R["V2 Nch=2 B=0.0 updn exact"][:checks][:box]; title = "Nch=2 (↑↓) exact"))
```

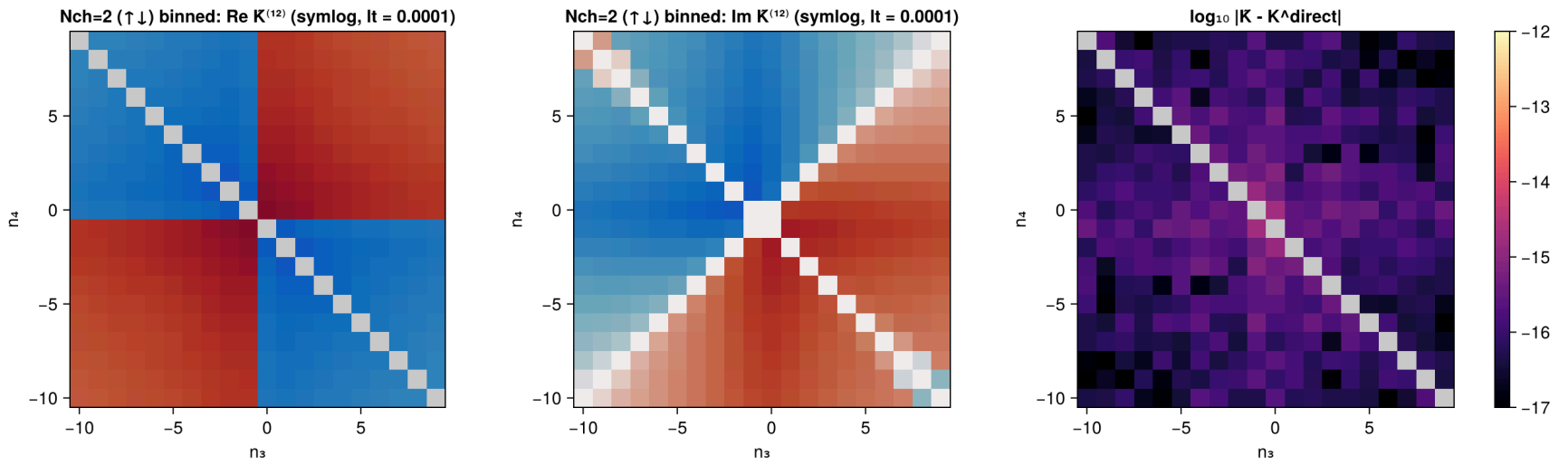
Out[19]:



In [20]:

```
savefig("chain_K_binned", K_heatmaps(R["V2 Nch=2 B=0.0 updn log"][:checks][:box]; title = "Nch=2 (↑↓) binned"))
```

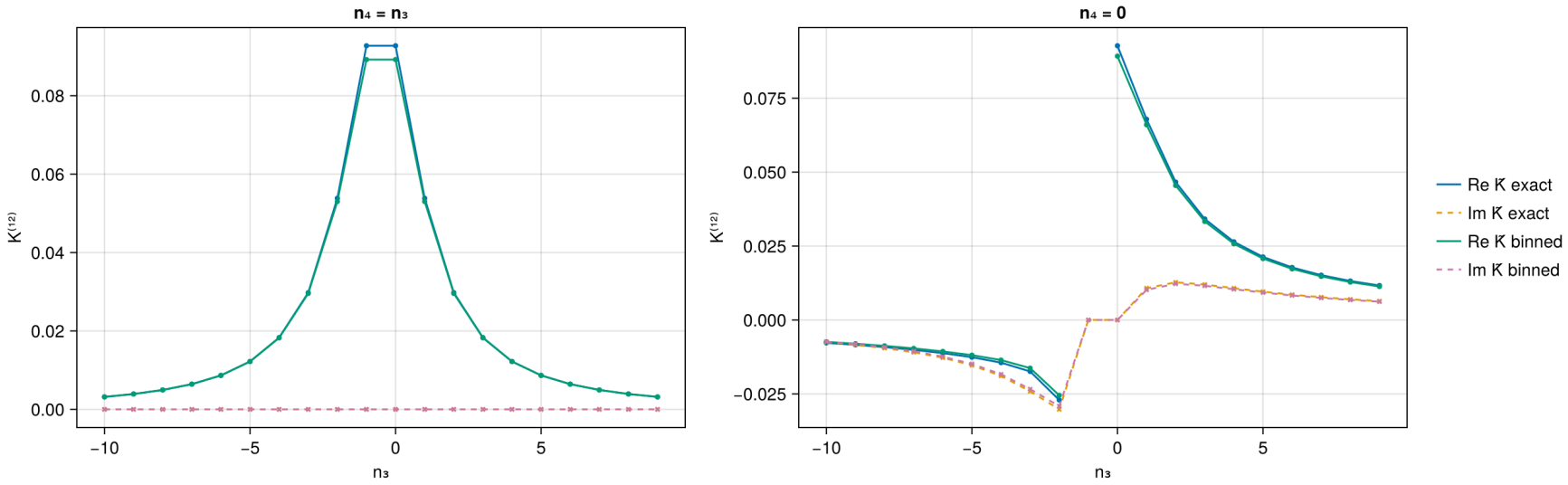
Out[20]:



In [21]:

```
# two frequency slices of K~ vs K^direct
function K_slices(keys; labels = keys)
    fig = Figure(size = (1300, 420))
    for (j, (sl, ttl)) in enumerate(((:diag, "n₄ = n₃"), (:fix, "n₄ = 0")))
        ax = Axis(fig[1, j]; xlabel = "n₃", ylabel = "K⁽¹²⁾", title = ttl)
        for (k, lb) in zip(keys, labels)
            bx = R[k][:checks][:box]
            idx = sl === :diag ? [(i, i) for i in eachindex(bx.n3s)] : [(i, findfirst(==(0), bx.n4s)) for i in eachindex(bx.n3s)]
            lines!(ax, bx.n3s, [real(bx.Kt[i...]) for i in idx]; label = "Re K̃ $lb")
            scatter!(ax, bx.n3s, [real(bx.Kd[i...]) for i in idx]; markersize = 6, marker = :circle)
            lines!(ax, bx.n3s, [imag(bx.Kt[i...]) for i in idx]; linestyle = :dash, label = "Im K̃ $lb")
            scatter!(ax, bx.n3s, [imag(bx.Kd[i...]) for i in idx]; markersize = 6, marker = :xcross)
        end
        j == 2 && Legend(fig[1, 3], ax; framevisible = false)
    end
    fig
end
savefig("chain_K_slices", K_slices(["V2 Nch=2 B=0.0 updn exact", "V2 Nch=2 B=0.0 updn log"]; labels = ["exact", "binned"]))
```

Out[21]:

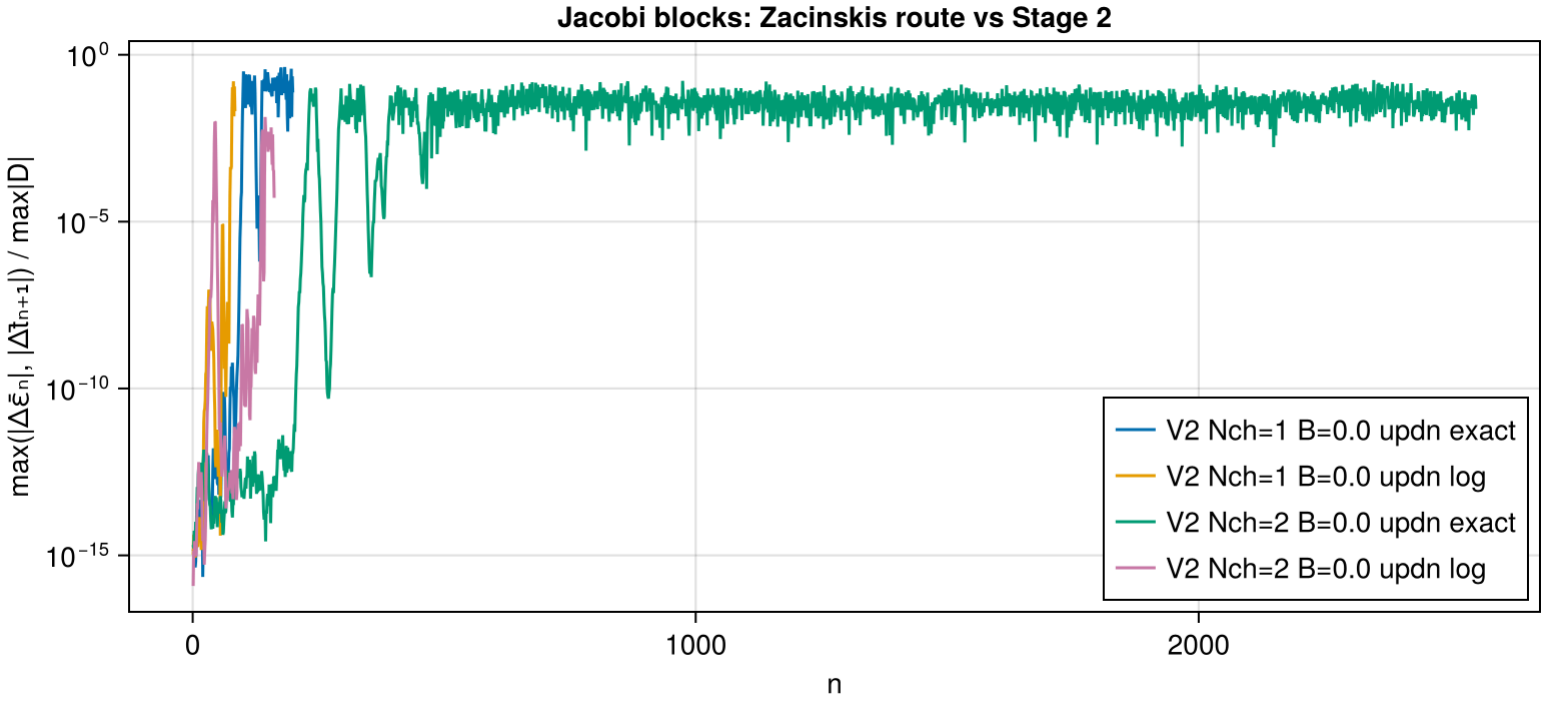


In [22]:

```
# Zacinskis cross-check (C2, [VD] Prop. 11)
@printf("%-30s %9s %9s %9s %9s %9s %7s %7s %7s\n", "case", "S-Ñ²", "Ũ₁", "spectra", "weights", "Σ^S", "L_zac", "L", "n_agree")
for k in vcat(["V2 Nch=$N B=0.0 $c $g" for N in (1, 2) for c in (:updn, :upup) for g in (:exact, :log)], ["V1 atom T=0.1 updn"])
    z = R[k][:checks][:zac]
    @printf("%-30s %9.1e %9.1e %9.1e %9.1e %9.1e %7d %7d %7d\n", k, z.S_err, z.U1_err, z.spec_err, z.weight_err, z.sigma_err, z.L_zac, z.L_chain, z.n_agree)
end
fig = Figure(size = (800, 380))
ax = Axis(fig[1, 1]; yscale = log10, xlabel = "n", ylabel = "max(|Δε̃ₙ|, |Δt̃ₙ₊₁|) / max|D|", title = "Jacobi blocks: Zacinskis route vs Stage 2")
for k in ("V2 Nch=1 B=0.0 updn exact", "V2 Nch=1 B=0.0 updn log", "V2 Nch=2 B=0.0 updn exact", "V2 Nch=2 B=0.0 updn log")
    z = R[k][:checks][:zac]
    lines!(ax, eachindex(z.block_dev), fl.(z.block_dev); label = k)
end
axislegend(ax; position = :rb)
savefig("zacinskis_blocks", fig)
```

case	S-Ñ²	Ũ₁	spectra	weights	Σ^S	L_zac	L	n_agree
V2 Nch=1 B=0.0 updn exact	8.0e-16	8.3e-16	3.2e-15	1.2e-13	3.3e-14	199	203	91
V2 Nch=1 B=0.0 updn log	1.1e-16	2.8e-16	4.5e-15	5.7e-14	2.2e-14	84	86	28
V2 Nch=1 B=0.0 upup exact	8.0e-16	8.3e-16	2.9e-15	1.1e-13	3.5e-14	201	203	91
V2 Nch=1 B=0.0 upup log	1.1e-16	2.8e-16	4.5e-15	5.7e-14	2.2e-14	84	86	28
V2 Nch=2 B=0.0 updn exact	3.5e-16	1.4e-16	5.6e-15	1.6e-13	5.8e-14	2551	2612	209
V2 Nch=2 B=0.0 updn log	1.2e-16	3.6e-16	3.6e-15	2.6e-14	9.1e-15	162	162	33
V2 Nch=2 B=0.0 upup exact	3.5e-16	1.4e-16	1.6e-14	1.2e-12	4.6e-13	2544	2612	210
V2 Nch=2 B=0.0 upup log	1.2e-16	3.6e-16	3.6e-15	2.6e-14	9.1e-15	162	162	33
V1 atom T=0.1 updn	1.1e-16	3.3e-16	4.0e-16	2.8e-14	1.2e-15	1	1	1

Out[22]:



4. Untruncated NRG (V3a)

The same chains with `NRGParams(N = Nch, Nkeep = typemax(Int))`: without truncation the Anders–Schiller basis is the exact eigenbasis, so NRG data must equal the binned ED data of V2b (mpNRG V3c accuracy, about 10^{-14}), and all guaranteed checks must hold. `DiscreteData` from NRG and from ED are compared directly.

In [23]:

```
@printf("%-30s %9s %9s %9s\n", "case", "max|ΔW̃|", "max|Δw^B|", "max|Δs|")
for N in (1, 2), f in (0.0, 0.05), c in (:updn, :upup)
    k = "V3a Nch=$N B=$f $c"; df = R[k][:diff]
    @printf("%-30s %9.1e %9.1e %9.1e\n", k, df.W, df.wB, df.s)
end
println()
chain_table(["V3a Nch=$N B=$f $c" for N in (1, 2) for f in (0.0, 0.05) for c in (:updn, :upup)])
```

```
case                           max|ΔW̃| max|Δw^B|   max|Δs|
V3a Nch=1 B=0.0 updn           4.4e-15   3.6e-16   2.8e-15
```

```
V3a Nch=1 B=0.0 upup            4.4e-15   5.8e-16   5.2e-16
V3a Nch=1 B=0.05 updn           1.7e-15   5.8e-16   8.6e-16
V3a Nch=1 B=0.05 upup           1.7e-15   3.6e-16   3.5e-16
V3a Nch=2 B=0.0 updn            9.2e-15   9.4e-16   4.7e-15
V3a Nch=2 B=0.0 upup            9.2e-15   7.9e-16   1.4e-15
V3a Nch=2 B=0.05 updn           5.3e-15   1.1e-15   3.2e-15
V3a Nch=2 B=0.05 upup           5.3e-15   3.4e-16   7.7e-16

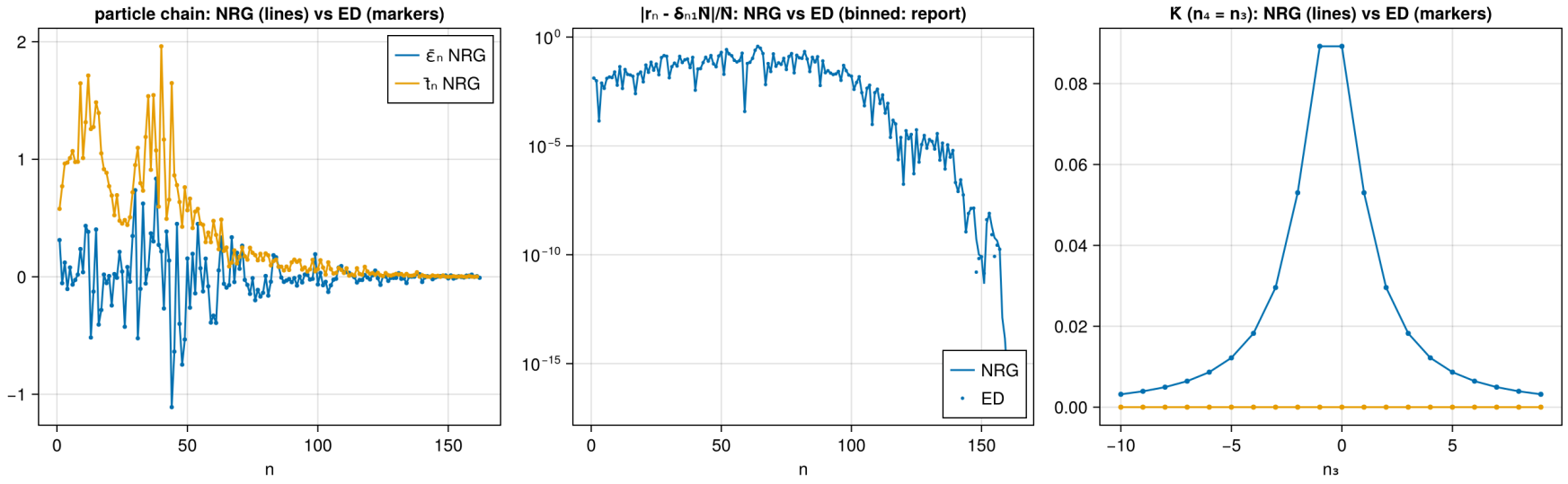
case                          bins     R    L  L_B        C1        C2        C3        C4   C4 unst        C5        C6        C7        C8     t [s]
V3a Nch=1 B=0.0 updn           195    87   86   55   2.0e-15   7.0e-14   2.0e-01   3.7e-01         –   0.0e+00   5.2e-16   5.8e-16   3.1e-15      25.5
V3a Nch=1 B=0.0 upup           195    87   86   55   2.0e-15   7.0e-14   2.0e-01   3.7e-01         –   0.0e+00   7.2e-16   5.0e-16   3.3e-14       1.0
V3a Nch=1 B=0.05 updn          195   102  101   57   2.6e-15   3.6e-14   2.4e-01   5.7e-01         –   0.0e+00   7.1e-16   2.4e-14   2.8e-14       1.2
V3a Nch=1 B=0.05 upup          195   102  101   57   2.6e-15   3.6e-14   2.4e-01   5.7e-01         –   0.0e+00   7.2e-16   3.4e-14   1.8e-13       1.2
V3a Nch=2 B=0.0 updn           195   163  162   91   1.7e-15   3.2e-14   4.5e-01   3.8e-01         –   0.0e+00   4.4e-16   5.7e-16   2.1e-14       6.1
V3a Nch=2 B=0.0 upup           195   163  162   91   1.7e-15   3.2e-14   4.5e-01   3.8e-01         –   0.0e+00   4.3e-16   6.2e-16   3.9e-14       5.7
V3a Nch=2 B=0.05 updn          195   175  174   91   1.8e-15   5.2e-14   3.6e-01   3.4e-01         –   0.0e+00   6.5e-16   5.0e-16   2.7e-14       6.2
V3a Nch=2 B=0.05 upup          195   175  174   91   1.8e-15   5.2e-14   3.6e-01   3.4e-01         –   0.0e+00   7.3e-16   6.1e-16   2.7e-14       6.3
```

In [24]:

```
fig = Figure(size = (1300, 420))
o = R["V3a Nch=2 B=0.0 updn"]; e = R["V2 Nch=2 B=0.0 updn log"]
ch = o[:res].ch; che = e[:res].ch
ax = Axis(fig[1, 1]; xlabel = "n", title = "particle chain: NRG (lines) vs ED (markers)")
lines!(ax, [real(x[1]) for x in ch.Jbar.eps]; label = "ε̃ₙ NRG"); scatter!(ax, [real(x[1]) for x in che.Jbar.eps]; markersize = 5)
lines!(ax, [real(x[1]) for x in ch.Jbar.lower]; label = "t̃ₙ NRG"); scatter!(ax, [real(x[1]) for x in che.Jbar.lower]; markersize = 5)
axislegend(ax)
ax = Axis(fig[1, 2]; yscale = log10, xlabel = "n", title = "|rₙ - δₙ₁Ñ|/Ñ: NRG vs ED (binned: report)")
lines!(ax, fl.(HA.leakage_profile(ch)); label = "NRG"); scatter!(ax, fl.(HA.leakage_profile(che)); markersize = 4, label = "ED")
axislegend(ax; position = :rb)
bx = o[:checks][:box]; bxe = e[:checks][:box]
ax = Axis(fig[1, 3]; xlabel = "n₃", title = "K̃ (n₄ = n₃): NRG (lines) vs ED (markers)")
lines!(ax, bx.n3s, [real(bx.Kt[i, i]) for i in eachindex(bx.n3s)]); scatter!(ax, bxe.n3s, [real(bxe.Kt[i, i]) for i in eachindex(bxe.n3s)]; markersize = 6)
lines!(ax, bx.n3s, [imag(bx.Kt[i, i]) for i in eachindex(bx.n3s)]); scatter!(ax, bxe.n3s, [imag(bxe.Kt[i, i]) for i in eachindex(bxe.n3s)]; markersize = 6)
m_ = .!isnan.(real.(bx.Kt))
@printf("max|K̃(NRG) - K̃(ED)| / max|K̃| = %.1e\n", maximum(abs, bx.Kt[m_] - bxe.Kt[m_]) / maximum(abs, bxe.Kt[m_]))
savefig("v3a_overlay", fig)
```

```
max|K̃(NRG) - K̃(ED)| / max|K̃| = 2.4e-14
```

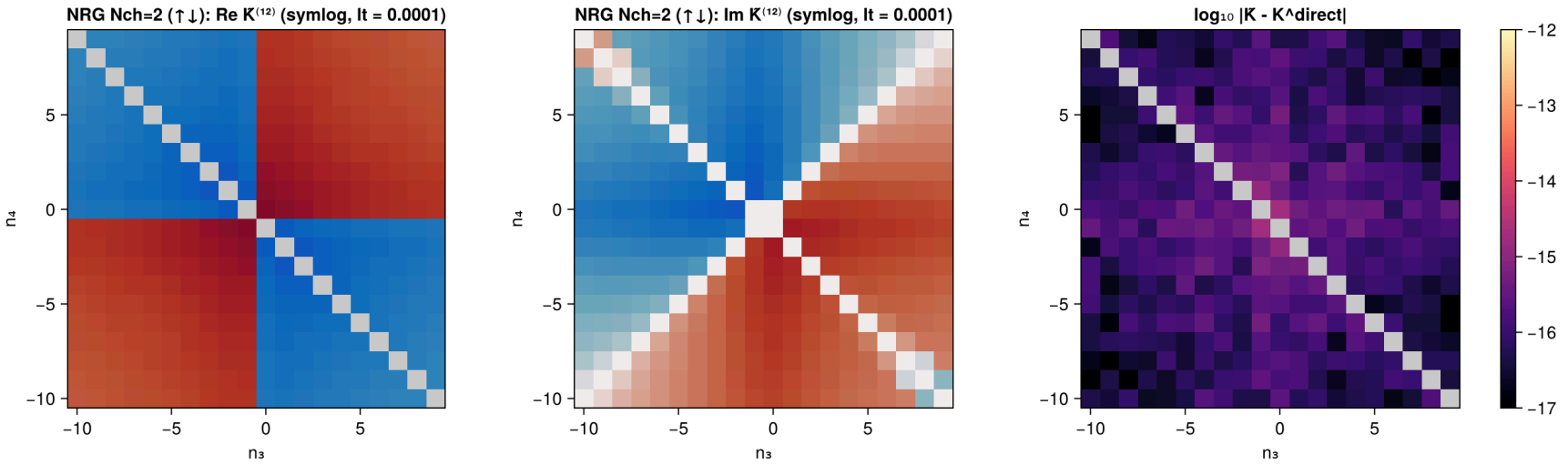
Out[24]:



In [25]:

```
savefig("v3a_K", K_heatmaps(R["V3a Nch=2 B=0.0 updn"][:checks][:box]; title = "NRG Nch=2 (↑↓)"))
```

Out[25]:



5. Decoupled tail, truncated NRG (V3b)

As mpNRG V4d: chain of length 3 with $t_n = \varepsilon_n = 0$ for $n > n_c = 1$, `Nkeep = 40`, $T = 0.05$. With the decoupled tail the Anders–Schiller states are exact eigenstates and the NRG approximation, the shell-diagonal expansion and the slicing are exact, so **truncated** NRG data equal ED of the same chain; this is the only regime in which truncated data are exact. Compared with ED of the same chain on the same grid. A failure here would be a bug, not NRG physics.

In [26]:

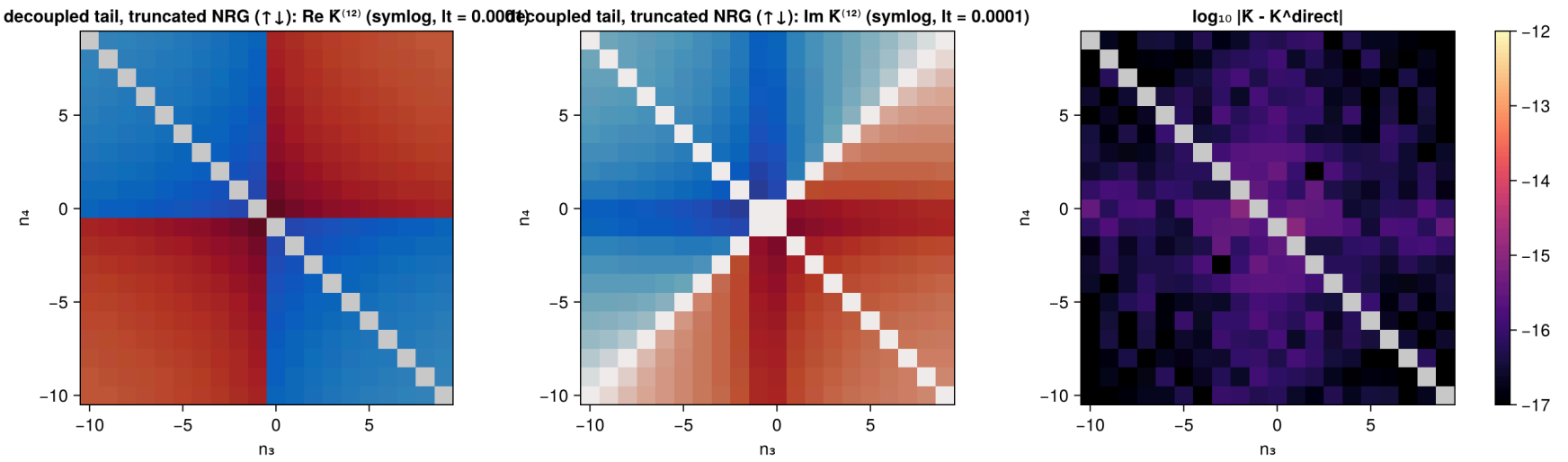
```
for c in (:updn, :upup)
    o = R["V3b $c"]; tr = o[:data].meta[:truncation]; df = o[:diff]
    @printf("%s: kept per shell %s, discarded %s; data vs ED: max|ΔW̃| = %.1e, max|Δw^B| = %.1e, max|Δs| = %.1e\n",
            chs(c), tr.kept, tr.disc, df.W, df.wB, df.s)
end
println()
chain_table(["V3b updn", "V3b upup"])
println("\nED of the same chain (binned, same grid):")
for c in (:updn, :upup)
    ce = R["V3b $c"][:ed][:checks]
    @printf("  %s: C1 %.1e  C5 %.1e  C6 %.1e  C7 %.1e  C8 %.1e; K̃(NRG) vs K̃(ED): %.1e\n", chs(c), ce[:C1], ce[:C5], ce[:C6], ce[:C7], ce[:C8],
            (bx = R["V3b $c"][:checks][:box]; bxe = ce[:box]; m_ = .!isnan.(real.(bx.Kt)); maximum(abs, bx.Kt[m_] - bxe.Kt[m_]) / maximum(abs, bxe.Kt[m_])))
end
```

```
(↑↓): kept per shell [16, 42, 40, 0], discarded [0, 22, 128, 160]; data vs ED: max|ΔW̃| = 3.2e-15, max|Δw^B| = 7.2e-16, max|Δs| = 2.8e-15
(↑↑): kept per shell [16, 42, 40, 0], discarded [0, 22, 128, 160]; data vs ED: max|ΔW̃| = 3.2e-15, max|Δw^B| = 5.4e-16, max|Δs| = 4.9e-16
case                              bins     R    L  L_B        C1        C2        C3        C4   C4 unst        C5        C6        C7        C8     t [s]
V3b updn                           195    87   86   55   1.2e-15   7.1e-14   2.0e-01   3.7e-01         –   0.0e+00   3.7e-16   4.9e-16   3.4e-15       2.7
V3b upup                           195    87   86   55   1.2e-15   7.1e-14   2.0e-01   3.7e-01         –   0.0e+00   1.1e-15   9.9e-16   3.1e-14       2.0

ED of the same chain (binned, same grid):
  (↑↓): C1 2.1e-15  C5 0.0e+00  C6 3.2e-16  C7 4.9e-16  C8 3.2e-15; K̃(NRG) vs K̃(ED): 2.9e-15
  (↑↑): C1 2.1e-15  C5 0.0e+00  C6 6.8e-16  C7 4.0e-16  C8 2.8e-14; K̃(NRG) vs K̃(ED): 2.7e-14
```

In [27]:
```
savefig("v3b_K", K_heatmaps(R["V3b updn"][:checks][:box]; title = "decoupled tail, truncated NRG (↑↓)"))
```

Out[27]:



6. Physical chain, truncated NRG (V3c)

`siam(U = 0.5, εd = -0.1, Δ = 0.1)` on the physical Wilson chain, $\Lambda = 4$, $z = 1$, length `chain_length_for(T; Λ = 4)` (as the mpNRG validation runs), $N_{\mathrm{keep}} \in \{64, 128, 256, 512\}$, $T \in \{10^{-2}, 10^{-4}\}$, both channels, on `LogGrid(emin = 1e-9, emax = 10, ndec = 16)` (deviation from the default `emin = 1e-5`, which clamps millions of contributions at $T = 10^{-4}$; see `vertex_krylov_STATUS.md`). Truncated NRG data are **not** exact data of a single generator, so nothing is guaranteed: every check is reported. The central question is how much the inconsistency of the NRG data, measured by the factorization residual C7, costs in the vertex, measured by C8 against the direct estimator from the same PSFs. C1 is algebraic (it holds for any Hermitian data); C3 uses the $U = 0$ propagator of the same chain from its single-particle Hamiltonian (`vk_G0_chain`); C4 has no guarantee for binned NRG data (Prop. 13 does not apply). Convergence in N_{keep} is measured against $N_{\mathrm{keep}} = 512$.

Truncated NRG weights violate two preconditions of Stage 1 that exact and binned ED data satisfy to rounding: the $2N \times 2N$ Pelz weights $\widetilde{\mathbf{W}}_\beta$ are not Hermitian bin by bin (their sum is), and their Hermitian part is not positive semidefinite. Stage 1 factors the Hermitian part and drops its negative eigenvalues (both reported below), so the factored measure has $v^\dagger v \neq 1$. Where the specified assertion $\|v^\dagger v - 1\| \leq 10^{-12}$ fails, Stage 1 runs with the opt-in variant `gram = :general` ($P = 1 - v(v^\dagger v)^{-1} v^\dagger$, $\sigma = (v^\dagger v)^{-1} v^\dagger w$

; identical to the specification for $v^\dagger v = 1$; deviation in `vertex_krylov_STATUS.md`). All checks compare with the raw data, so C1 and C8 include the cost of what Stage 1 drops.

In [28]:

```
V3CK = [(T, Nk, c) for T in V3C_TS for Nk in V3C_NKEEPS for c in (:updn, :upup)]
v3(T, Nk, c) = R["V3c T=$T Nk=$Nk $c"]
@printf("%-8s %6s %-5s %3s %5s %5s %5s %5s %4s %9s %9s %9s %9s %9s %9s %9s %8s %7s\n", "T", "Nkeep", "chan", "N", "kept",
        "bins", "R", "L", "L_B", "C1", "C2", "C3", "C4", "C5", "C6", "C7", "C8", "clamped", "t_NRG")
for (T, Nk, c) in V3CK
    o = v3(T, Nk, c); ck = o[:checks]; d = o[:data]
    @printf("%-8.0e %6d %-5s %3d %5d %5d %5d %5d %4d %9.1e %9.1e %9.1e %9.1e %9.1e %9.1e %9.1e %8d %7.0f\n", T, Nk, c,
            d.meta[:N], maximum(d.meta[:truncation].kept), count(β -> !iszero(d.W[β]), eachindex(d.b)), ck[:R], ck[:L], ck[:LB],
            ck[:C1], ck[:C2], ck[:C3], ck[:C4], ck[:C5], ck[:C6], ck[:C7], ck[:C8], d.meta[:nclamped], d.meta[:time_nrg] + d.meta[:time_data])
end
println("\nsanity (sum rules vs thermal_average_product, 3p totals, PSD), max over V3c:")
S = [v3(k...)[:checks][:sanity] for k in V3CK]
@printf("  herm %.1e, min eig/max‖W‖ %.1e, sum rule %.1e, 3p totals %.1e, C5 %.1e, Boltzmann-shortcut diff %.1e\n",
        maximum(s.herm_err for s in S), minimum(s.min_eig_rel for s in S), maximum(s.sumrule_err for s in S),
        maximum(s.totals3_err for s in S), maximum(s.C5 for s in S), maximum(s.boltzmann_diff for s in S))
println("C7 per family (max over V3c): ", Dict(f => @sprintf("%.1e", maximum(v3(k...)[:checks][:C7_family][f] for k in V3CK)) for f in 1:6))
```

```
T         Nkeep chan    N  kept  bins     R     L  L_B        C1        C2        C3        C4        C5        C6        C7        C8  clamped   t_NRG
1e-02        64 updn   15    66   214   393   392  251   1.3e-03   6.5e-14   5.6e-01   4.2e-01   0.0e+00   2.3e-16   1.4e-02   1.2e-01        0      41
1e-02        64 upup   15    66   214   393   392  251   1.3e-03   6.5e-14   5.6e-01   4.2e-01   0.0e+00   1.4e-15   2.1e-02   7.9e-02        0      13
1e-02       128 updn   15   130   226   429   425  257   4.4e-04   3.0e-14   3.9e-01   4.3e-01   0.0e+00   2.0e-15   3.1e-03   5.4e-04        0      38
1e-02       128 upup   15   130   226   429   425  257   4.4e-04   3.0e-14   3.9e-01   4.3e-01   0.0e+00   1.8e-15   2.3e-03   4.6e-04        0      38
1e-02       256 updn   15   257   245   468   467  269   2.3e-06   4.7e-13   2.5e-01   5.8e-01   0.0e+00   1.5e-15   5.1e-08   9.8e-05     6108     171
1e-02       256 upup   15   257   245   468   467  269   2.3e-06   4.7e-13   2.5e-01   5.8e-01   0.0e+00   1.6e-15   2.9e-08   1.5e-04     6108     170
1e-02       512 updn   15   514   254   492   490  281   2.2e-06   5.8e-13   2.8e-01   5.4e-01   0.0e+00   2.1e-15   4.7e-08   8.9e-05    14528     686
1e-02       512 upup   15   514   254   492   490  281   2.2e-06   5.8e-13   2.8e-01   5.4e-01   0.0e+00   1.2e-15   2.4e-08   1.2e-04    14528     695
1e-04        64 updn   22    66   268   481   479  272   1.4e-02   1.3e-13   6.0e-01   5.0e-01   0.0e+00   1.4e-05   1.5e-02   3.7e-01  1052244      27
1e-04        64 upup   22    66   268   481   479  272   1.4e-02   1.3e-13   6.0e-01   5.0e-01   0.0e+00   1.4e-05   2.3e-02   5.8e-01  1052244      28
1e-04       128 updn   22   130   286   532   529  276   4.4e-04   2.2e-14   4.2e-01   5.7e-01   0.0e+00   6.4e-07   2.4e-03   3.1e-03  3788960      74
1e-04       128 upup   22   130   286   532   529  276   4.4e-04   2.2e-14   4.2e-01   5.7e-01   0.0e+00   6.4e-07   2.4e-03   2.8e-03  3788960      75
1e-04       256 updn   22   257   292   556   546  280   3.4e-05   3.4e-14   2.6e-01   6.4e-01   0.0e+00   2.6e-06   2.5e-04   7.3e-04 13731404     386
1e-04       256 upup   22   257   292   556   546  280   3.4e-05   3.4e-14   2.6e-01   6.4e-01   0.0e+00   2.6e-06   2.3e-04   5.7e-04 13731404     381
1e-04       512 updn   22   514   307   599   596  280   1.1e-04   9.2e-14   2.8e-01   4.6e-01   0.0e+00   2.8e-06   2.3e-04   7.0e-04 36512928    1506
1e-04       512 upup   22   514   307   599   596  280   1.1e-04   9.2e-14   2.8e-01   4.6e-01   0.0e+00   2.8e-06   2.0e-04   4.5e-04 36512928    1530

sanity (sum rules vs thermal_average_product, 3p totals, PSD), max over V3c:
  herm 1.4e-02, min eig/max‖W‖ -4.8e-03, sum rule 1.1e-15, 3p totals 4.2e-15, C5 0.0e+00, Boltzmann-shortcut diff 1.9e+06
C7 per family (max over V3c): Dict
(3 => "1.5e-02", 2 => "2.4e-03", 1 => "2.3e-02", 4 => "3.7e-03", 5 => "3.7e-03", 6 => "1.8e-02")
```

In [29]:

```
# properties of the truncated NRG data that Stage 1 meets (report): W̃_β bin by bin is not Hermitian (Stage 1
# factors the Hermitian part) and its Hermitian part is not PSD (Stage 1 drops the negative eigenvalues), so the
# factored measure has v†v ≠ 1; where ‖v†v - 1‖ > 1e-12 Stage 1 runs with gram = :general (deviation, see STATUS);
# negative bosonic weights are clamped to 0
@printf("%-8s %6s %-5s %12s %12s %10s %8s %11s %8s %15s\n", "T", "Nkeep", "chan", "antiherm/‖W‖", "neg. mass", "worst neg",
        "#bins<0", "‖v†v - 1‖", "general", "w^B<0: n, min")
for (T, Nk, c) in V3CK
    p = v3(T, Nk, c)[:psd]
    @printf("%-8.0e %6d %-5s %12.1e %12.1e %10.1e %8d %11.1e %8s %15s\n", T, Nk, c, p.antiherm_rel, p.neg_mass, p.worst_rel,
            p.nbins_neg, p.norm_err, p.gram_general, @sprintf("%d, %.1e", p.wB_neg...))
end
```

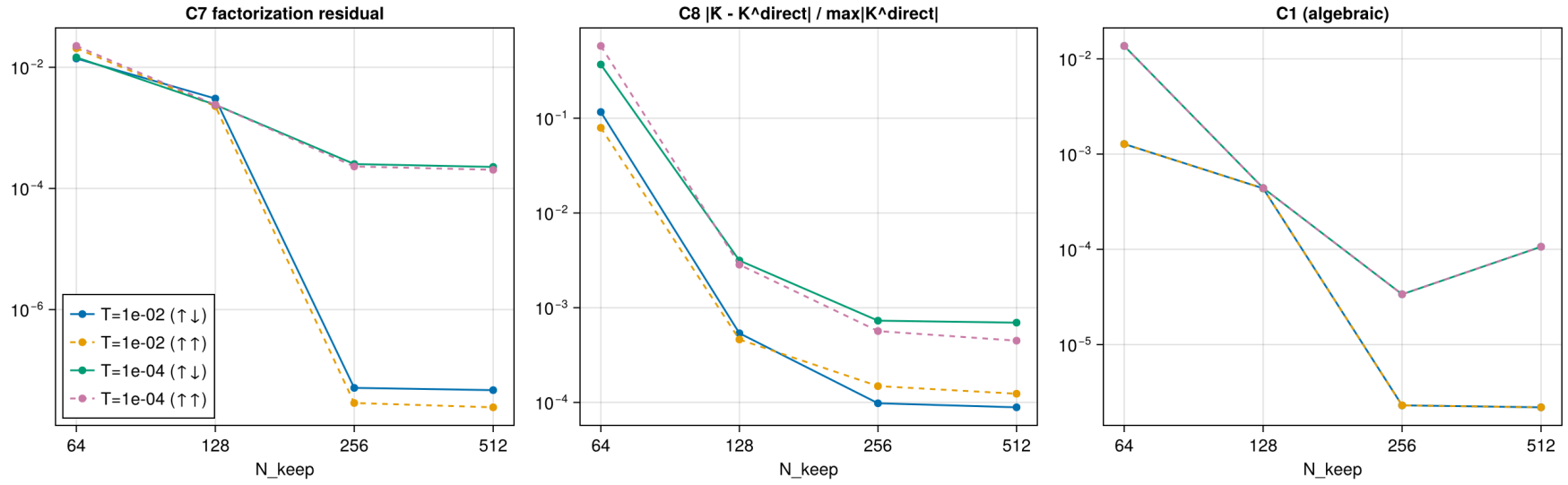
```
T         Nkeep chan  antiherm/‖W‖    neg. mass  worst neg  #bins<0   ‖v†v - 1‖  general   w^B<0: n, min
1e-02        64 updn       7.0e-03      1.7e-04    3.4e-01       15     1.3e-04     true      0, 0.0e+00
```

```
1e-02       64 upup       7.0e-03      1.7e-04    3.4e-01      15     1.3e-04     true      0, 0.0e+00
1e-02      128 updn       3.1e-03      1.0e-05    2.8e-03      10     2.2e-06     true      0, 0.0e+00
1e-02      128 upup       3.1e-03      1.0e-05    2.8e-03      10     2.2e-06     true      0, 0.0e+00
1e-02      256 updn       2.2e-03      2.1e-13    2.5e-03       6     1.8e-13    false      0, 0.0e+00
1e-02      256 upup       2.2e-03      2.1e-13    2.5e-03       6     1.8e-13    false      0, 0.0e+00
1e-02      512 updn       1.0e-03      2.2e-13    1.4e-03       5     1.9e-13    false      0, 0.0e+00
1e-02      512 upup       1.0e-03      2.2e-13    1.4e-03       5     1.9e-13    false      0, 0.0e+00
1e-04       64 updn       6.5e-03      9.1e-04    1.0e+00      40     7.3e-04     true     7, -1.1e-05
1e-04       64 upup       6.5e-03      9.1e-04    1.0e+00      40     7.3e-04     true     7, -1.1e-05
1e-04      128 updn       3.2e-03      1.5e-05    1.0e-01      26     2.6e-06     true     3, -5.0e-07
1e-04      128 upup       3.2e-03      1.5e-05    1.0e-01      26     2.6e-06     true     3, -5.0e-07
1e-04      256 updn       3.4e-03      2.4e-06    1.0e+00      18     2.1e-06     true     1, -3.4e-06
1e-04      256 upup       3.4e-03      2.4e-06    1.0e+00      18     2.1e-06     true     1, -3.4e-06
1e-04      512 updn       2.1e-03      1.3e-06    1.0e+00       9     6.0e-07     true     1, -3.6e-06
1e-04      512 upup       2.1e-03      1.3e-06    1.0e+00       9     6.0e-07     true     1, -3.6e-06
```

In [30]:

```julia
# C7 (data inconsistency) and C8 (cost in the vertex) against Nkeep and T
fig = Figure(size = (1300, 420))
for (j, (key, ttl)) in enumerate(((:C7, "C7 factorization residual"), (:C8, "C8 |K̃ - K^direct| / max|K^direct|"),
                                  (:C1, "C1 (algebraic)")))
    ax = Axis(fig[1, j]; xscale = log2, yscale = log10, xlabel = "N_keep", title = ttl, xticks = collect(V3C_NKEEPS))
    for T in V3C_TS, c in (:updn, :upup)
        scatterlines!(ax, collect(V3C_NKEEPS), [fl(v3(T, Nk, c)[:checks][key]) for Nk in V3C_NKEEPS];
                      label = @sprintf("T=%.0e %s", T, chs(c)), linestyle = c === :updn ? :solid : :dash)
    end
    j == 1 && axislegend(ax; position = :lb)
end
savefig("v3c_C7_C8", fig)
```

Out[30]:



In [31]:

```julia
# convergence of K̃ and of K^direct in Nkeep (relative to Nkeep = 512, over the Matsubara box)
println("max|K(Nkeep) - K(512)| / max|K(512)|:")
@printf("%-8s %-5s %s\n", "T", "chan", join([@sprintf("%15s", "Nkeep=$Nk (K̃, K^dir)") for Nk in V3C_NKEEPS[1:end-1]], " "))
for T in V3C_TS, c in (:updn, :upup)
    ref = v3(T, 512, c)[:checks][:box]; m_ = .!isnan.(real.(ref.Kt))
    vals = map(V3C_NKEEPS[1:end-1]) do Nk
        bx = v3(T, Nk, c)[:checks][:box]
        @sprintf("%7.1e, %7.1e", maximum(abs, bx.Kt[m_] - ref.Kt[m_]) / maximum(abs, ref.Kt[m_]),
                 maximum(abs, bx.Kd[m_] - ref.Kd[m_]) / maximum(abs, ref.Kd[m_]))
    end
```

```
    @printf("%-8.0e %-5s %s\n", T, c, join([@sprintf("%15s", v) for v in vals], " "))
end
```

```
max|K(Nkeep) - K(512)| / max|K(512)|:
T        chan  Nkeep=64 (K̃, K^dir) Nkeep=128 (K̃, K^dir) Nkeep=256 (K̃, K^dir)
1e-02    updn  9.4e-02, 2.6e-02 1.4e-02, 1.4e-02 5.5e-03, 5.5e-03
1e-02    upup  5.9e-02, 2.0e-02 2.3e-02, 2.3e-02 6.8e-03, 6.7e-03
1e-04    updn  3.7e-01, 6.7e-01 5.3e-03, 5.6e-03 2.3e-03, 2.3e-03
1e-04    upup  4.3e-01, 4.5e-01 5.9e-03, 6.6e-03 2.1e-03, 2.1e-03
```

In [32]:

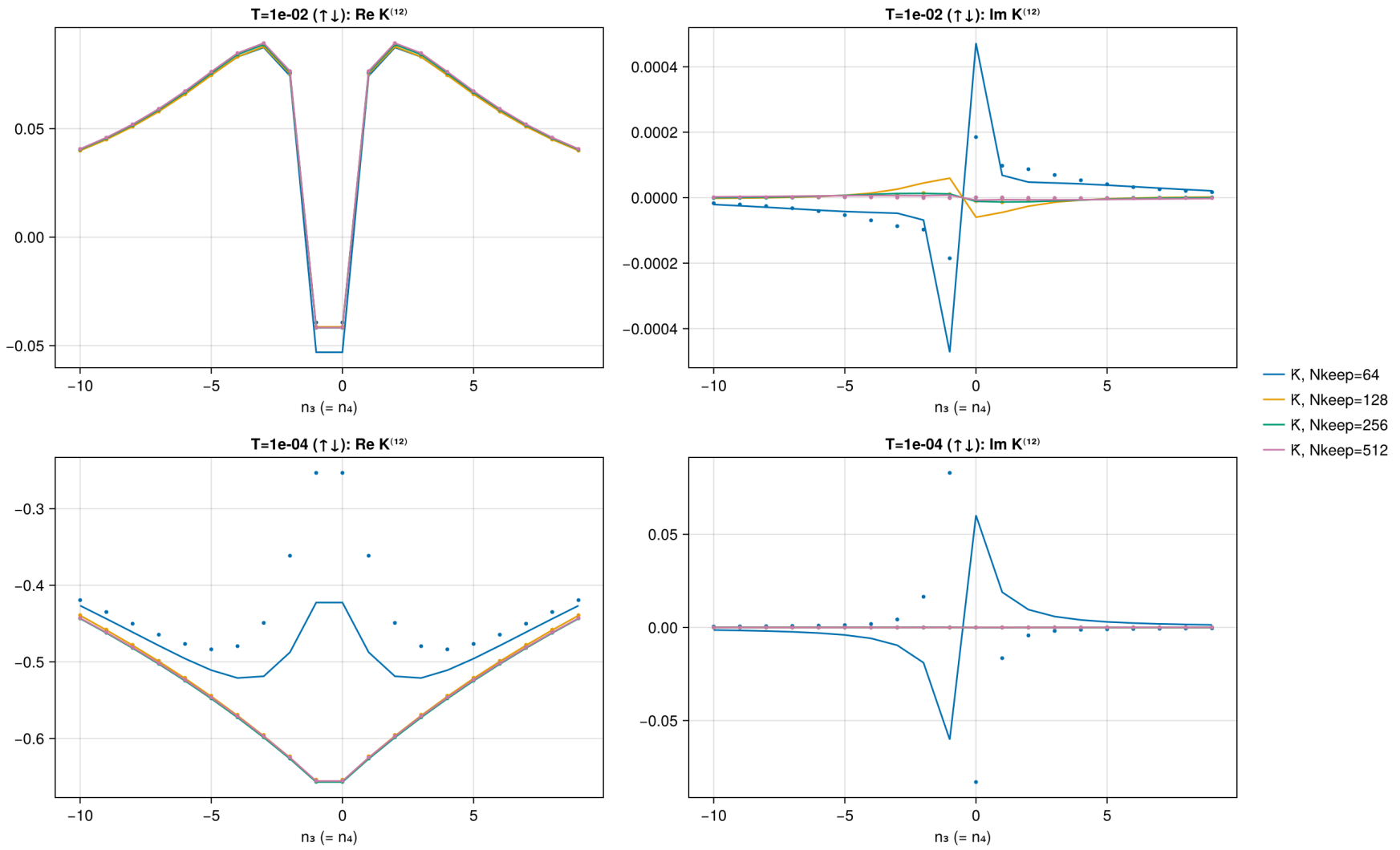
```
# particle-chain lengths and residuals, Float64 vs BigFloat (precision 128)
@printf("%-8s %6s %-5s %5s %5s %11s %11s %9s %9s %9s %8s\n", "T", "Nkeep", "chan", "L", "L_big", "resid f64", "resid big",
        "orth f64", "orth big", "C1 big", "n_agree")
for (T, Nk, c) in V3CK
    o = v3(T, Nk, c); ck = o[:checks]; b = o[:big]
    pd = o[:res] === nothing ? nothing : o[:res].ch.diag[:particle]
    @printf("%-8.0e %6d %-5s %5d %5d %11.1e %11.1e %9.1e %9.1e %9.1e %8d\n", T, Nk, c, ck[:L], b.L, ck[:particle_resid],
            Float64(b.resid), pd === nothing ? NaN : pd[:orth_loss], b.orth, b.C1, b.n_agree)
end
```

```
T         Nkeep chan      L L_big   resid f64   resid big  orth f64  orth big    C1 big  n_agree
1e-02        64 updn    392   392     1.0e-37     1.6e-82   1.7e-15   9.4e-38   1.3e-03       36
1e-02        64 upup    392   392     1.0e-37     1.6e-82   1.7e-15   9.4e-38   1.3e-03       36
1e-02       128 updn    425   425     2.4e-18     2.4e-18   1.6e-15   6.2e-38   4.4e-04      425
1e-02       128 upup    425   425     2.4e-18     2.4e-18   1.6e-15   6.2e-38   4.4e-04      425
1e-02       256 updn    467   467     4.9e-39     3.1e-49   2.0e-15   2.6e-36   2.3e-06      467
1e-02       256 upup    467   467     4.9e-39     3.1e-49   2.0e-15   2.6e-36   2.3e-06      467
1e-02       512 updn    490   490     3.5e-23     2.6e-33   1.8e-15   3.9e-37   2.2e-06      490
1e-02       512 upup    490   490     3.5e-23     2.6e-33   1.8e-15   3.9e-37   2.2e-06      490
1e-04        64 updn    479   479     6.7e-14     6.7e-14   1.7e-15   7.6e-38   1.4e-02       36
1e-04        64 upup    479   479     6.7e-14     6.7e-14   1.7e-15   7.6e-38   1.4e-02       36
1e-04       128 updn    529   529     2.5e-12     2.5e-12   1.6e-15   1.1e-37   4.4e-04      529
1e-04       128 upup    529   529     2.5e-12     2.5e-12   1.6e-15   1.1e-37   4.4e-04      529
1e-04       256 updn    546   546     1.2e-12     1.2e-12   1.6e-15   6.8e-38   3.4e-05      546
1e-04       256 upup    546   546     1.2e-12     1.2e-12   1.6e-15   6.8e-38   3.4e-05      546
1e-04       512 updn    596   596     7.8e-13     7.8e-13   2.4e-15   8.2e-38   1.1e-04      596
1e-04       512 upup    596   596     7.8e-13     7.8e-13   2.4e-15   8.2e-38   1.1e-04      596
```

In [33]:

```
# K̃ slices (n4 = n3) for all Nkeep, with K^direct (markers), channel (↑↓)
fig = Figure(size = (1300, 800))
for (i, T) in enumerate(V3C_TS), (j, part) in enumerate((real, imag))
    ax = Axis(fig[i, j]; xlabel = "n₃ (= n₄)", title = @sprintf("T=%.0e (↑↓): %s K⁽¹²⁾", T, part === real ? "Re" : "Im"))
    for Nk in V3C_NKEEPS
        bx = v3(T, Nk, :updn)[:checks][:box]
        lines!(ax, bx.n3s, [part(bx.Kt[k, k]) for k in eachindex(bx.n3s)]; label = "K̃, Nkeep=$Nk")
        scatter!(ax, bx.n3s, [part(bx.Kd[k, k]) for k in eachindex(bx.n3s)]; markersize = 5)
    end
    i == 1 && j == 2 && Legend(fig[1:2, 3], ax; framevisible = false)
end
savefig("v3c_K_slices", fig)
```

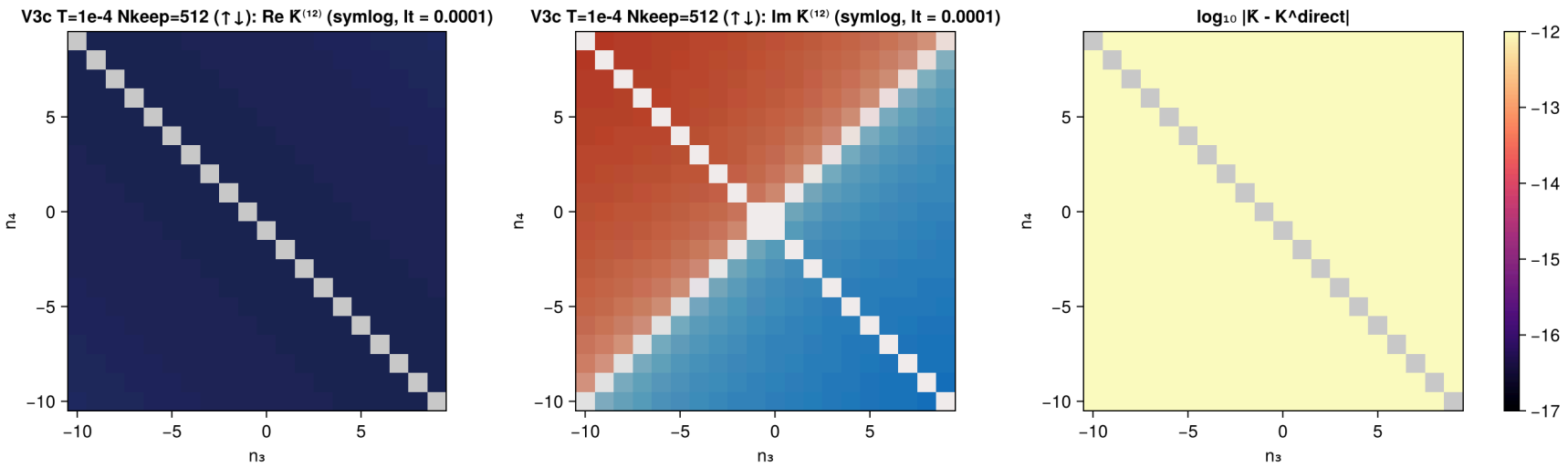
Out[33]:



In [34]:

```
savefig("v3c_K", K_heatmaps(v3(1e-4, 512, :updn)[:checks][:box]; title = "V3c T=1e-4 Nkeep=512 (↑↓)"))
```

Out[34]:



In [35]:

```
# Zacinskis cross-check on V3c data (C2 invariants; Jacobi blocks: n_agree)
@printf("%-26s %9s %9s %9s %9s %9s %7s %7s %7s\n", "case", "S-Ñ²", "Ũ₁", "spectra", "weights", "Σ^S", "L_zac", "L", "n_agree")
for (T, Nk, c) in V3CK
    z = v3(T, Nk, c)[:checks][:zac]
    @printf("%-26s %9.1e %9.1e %9.1e %9.1e %9.1e %7d %7d %7d\n", @sprintf("T=%.0e Nk=%d %s", T, Nk, c), z.S_err, z.U1_err,
            z.spec_err, z.weight_err, z.sigma_err, z.L_zac, z.L_chain, z.n_agree)
end
```

case	S-Ñ²	Ũ₁	spectra	weights	Σ^S	L_zac	L	n_agree
T=1e-02 Nk=64 updn	9.4e-16	2.1e-16	4.5e-15	6.5e-14	1.9e-14	392	392	32
T=1e-02 Nk=64 upup	9.4e-16	2.1e-16	4.5e-15	6.5e-14	1.9e-14	392	392	32
T=1e-02 Nk=128 updn	2.4e-16	1.1e-16	2.9e-15	3.0e-14	1.3e-14	425	425	33
T=1e-02 Nk=128 upup	2.4e-16	1.1e-16	2.9e-15	3.0e-14	1.3e-14	425	425	33
T=1e-02 Nk=256 updn	9.4e-16	4.4e-14	3.4e-15	4.7e-13	8.6e-14	467	467	33
T=1e-02 Nk=256 upup	9.4e-16	4.4e-14	3.4e-15	4.7e-13	8.6e-14	467	467	33
T=1e-02 Nk=512 updn	2.4e-16	4.7e-14	5.8e-15	5.8e-13	1.0e-13	490	490	33
T=1e-02 Nk=512 upup	2.4e-16	4.7e-14	5.8e-15	5.8e-13	1.0e-13	490	490	33
T=1e-04 Nk=64 updn	0.0e+00	1.4e-16	4.0e-15	1.3e-13	4.7e-14	478	479	32
T=1e-04 Nk=64 upup	0.0e+00	1.4e-16	4.0e-15	1.3e-13	4.7e-14	478	479	32
T=1e-04 Nk=128 updn	4.7e-16	1.1e-16	7.1e-15	2.2e-14	9.5e-15	531	529	32
T=1e-04 Nk=128 upup	4.7e-16	1.1e-16	7.1e-15	2.2e-14	9.5e-15	531	529	32
T=1e-04 Nk=256 updn	8.3e-16	2.2e-16	2.8e-15	3.4e-14	9.8e-15	547	546	33
T=1e-04 Nk=256 upup	8.3e-16	2.2e-16	2.8e-15	3.4e-14	9.8e-15	547	546	33
T=1e-04 Nk=512 updn	3.5e-16	1.7e-16	3.8e-15	9.2e-14	1.7e-14	596	596	34
T=1e-04 Nk=512 upup	3.5e-16	1.7e-16	3.8e-15	9.2e-14	1.7e-14	596	596	34

7. Findings

The numbers behind each finding are computed in the cell below the text.

F1 — the unstabilized particle chain leaks into the single-particle subspace (C4 report, amendment A1). The particle chain of Algorithm 2 with $\Pi = 1 - vv^\dagger$ keeps running after the cyclic space of $Qq^\dagger$ is exhausted and enters the single-particle invariant

subspace $\mathbb{K}_0 \ominus d^\dagger$ ([VD] Lemma 12), where $r_n = \bar{U}_n^\dagger D v$ is $O(1)$. Step 1 classified this as the failure of a guaranteed check; on review it is a mis-specification of the task sheet: for exact data with a bath C4 is guaranteed only for the stabilized chain of Prop. 13, which is now the default output of Stage 2 (amendments A1, A2). Evidence (cell below and Sec. 3):

- the leak $|r_n - \delta_{n1}\bar{N}|/\bar{N}$ starts at rounding level at $n = 1$ and grows roughly exponentially along the chain ("bath leakage" plot); the unstabilized chain is longer than the stabilized one by $\dim\Omega - 1$, the number of bath levels;
- BigFloat (128 bit) gives the same leak, so the seed is not rounding in the Lanczos recursion but the accuracy with which the Float64 ED data satisfy the equation of motion; the recursion amplifies this component until the residual exceeds the termination threshold `tol·max|D|` and the chain continues into the invariant subspace;
- with the stabilization ($\Pi = 1 - \Omega\Omega^\dagger$, $\dim\Omega$ = number of chain levels) C4 is at rounding level, and Σ^{S}, $\tilde{K}$ are unchanged; C1 and C8 of the unstabilized chain also hold: the spurious levels carry no weight in Σ^{S} or $\tilde{K}$.

Smallest reproducing case: `siam(U = 0.5, εd = -0.1, Δ = 0.1)`, chain sites $-1..0$ (16 states), $T = 0.05$, ExactGrid, channel (↑↓) (`V2 Nch=0`). Lemma 12 holds in exact arithmetic **with exact data**; Float64 data are exact only to rounding, which the unprojected invariant subspace does not forgive ([VD] Sec. 9: "such errors can grow along the chain").

F2 — C2: the Jacobi blocks of two Lanczos runs agree only on the leading blocks. The Zacinskis route (Ω' diagonalized, Lanczos from the image of $\bar{U}_1$) reproduces $S = \bar{N}^2$, the start block, the spectrum and first-block weights of PDP on $\mathrm{ran}\,P$, and Σ^{S} (C2 invariants, guaranteed under A3). Its Jacobi blocks agree with Stage 2 only up to `n_agree` and its chain may terminate at a different length: the measures have weights spanning $\sim e^{-|E|/T}$, many decades, and the late Jacobi coefficients of such measures are ill-conditioned (they are fixed by poles whose weights are near the termination threshold). Reported (A3).

F3 — $K^{(12)} \equiv 0$ in the atom's (↑↑) channel. $B = q_{\uparrow\uparrow} = U n_\downarrow$ commutes with H, $d_\uparrow$ and $q_\uparrow$, so all four correlators are independent of τ_1 and their regular parts vanish at $\omega_{12} \neq 0$; the six permutation terms are $O(1)$ and cancel in both $\tilde{K}$ and K^{direct} (also with operator-level `regular_sum`, `test/test_vertex_krylov.jl`). C8 as a ratio is then 0/0 and is measured on the scale of the permutation terms (deviation 4 in `vertex_krylov_STATUS.md`); the per-permutation agreement holds as everywhere.

F4 — V3c: truncated NRG data are not exact data of a single generator, and the vertex pays for it (report). Evidence in Sec. 6 and in the cell below.

- *Stage-1 preconditions fail.* The Pelz weights $\widetilde{\mathbf{W}}_\beta$ of truncated NRG are not Hermitian bin by bin, and their Hermitian part is not positive semidefinite. The specified algorithm therefore stops at its Stage-1 assertion $\|v^\dagger v - 1\| \leq 10^{-12}$ whenever the dropped

negative mass exceeds that level; Sec. 6 runs those cases with the reported variant `gram = :general`. The anti-Hermitian part shrinks with N_{keep} but does not vanish at $N_{\mathrm{keep}} = 512$; the negative part reaches rounding level at $T = 10^{-2}$ for $N_{\mathrm{keep}} \geq 256$ (where the specified Stage 1 runs) but not at $T = 10^{-4}$.

- *Checks that hold for any Hermitian data still hold:* the C2 invariants (Zacinskis route) and C5 stay at rounding level. C1 does not, because $\widetilde{G}$ is built from the raw weights while the chain sees only their Hermitian PSD part; C1 measures exactly what Stage 1 drops.
- *C7 vs C8.* The factorization residual C7 and the assembly error C8 both decrease with N_{keep}, but C8 is not controlled by C7 alone: at $T = 10^{-2}$ C7 drops to the 10^{-8} level for $N_{\mathrm{keep}} \geq 256$ while C8 levels off well above it. This is consistent with C8 being set at large N_{keep} by the parts of the data that Stage 1 drops (anti-Hermitian, non-PSD) rather than by the kernel factorization; this attribution is not isolated here.
- *Convergence in N_{keep}.* $\tilde{K}$ and K^{direct} change little between $N_{\mathrm{keep}} = 256$ and 512 compared with the change from 64 (convergence table in Sec. 6).
- *Report-only checks.* C3 (Dyson) and C4 (bath leakage) are of the same size as for the binned ED data of V2b, i.e. set by the binning; C6 is at rounding level at $T = 10^{-2}$ and limited at $T = 10^{-4}$ by the clamped negative bosonic weights.
- *Precision.* The BigFloat particle chain has the same length as the Float64 one, and the Jacobi blocks agree on all levels except at $N_{\mathrm{keep}} = 64$, where the data are the least consistent.

In [36]:

```
println("F1 — unstabilized (diagnostic) vs default (stabilized) particle chain, exact chain data")
@printf("%-28s %9s %11s %10s %6s %6s %4s %9s %9s %10s\n", "case", "C4 unst", "C4 big-uns", "C4 default", "L uns", "L def",
        "dimΩ", "C1 uns", "C8 uns", "n(>1e-9)")
for k in vcat(["V2 Nch=0 B=0.0 updn exact"], ["V2 Nch=$N B=$f $c exact" for N in (1, 2) for f in (0.0, 0.05) for c in (:updn, :upup)],
              ["V2 block N=2"])
    o = R[k]; u = o[:unstab]; ck = o[:checks]
    nfirst = findfirst(>(1e-9), u.profile)
    @printf("%-28s %9.2e %11s %10.1e %6d %6d %4d %9.1e %9.1e %10s\n", k, u.C4,
            haskey(o, :big_unstab) ? @sprintf("%.2e", o[:big_unstab].C4) : "—", ck[:C4], u.L, ck[:L], ck[:Omega_dim],
            u.C1, u.C8, nfirst === nothing ? "—" : string(nfirst))
end
zs = [R[k][:checks][:zac] for k in keys(R) if haskey(R[k], :checks) && haskey(R[k][:checks], :zac) && !startswith(k, "V3c")]
@printf("\nF2 — Steps 1 cases: C2 invariants max %.1e; n_agree / L from %.2f to %.2f (n_agree %d–%d)\n",
        maximum(R[k][:checks][:C2] for k in keys(R) if haskey(R[k], :checks) && !startswith(k, "V3c")),
        minimum(z.n_agree / z.L_chain for z in zs), maximum(z.n_agree / z.L_chain for z in zs),
        minimum(z.n_agree for z in zs), maximum(z.n_agree for z in zs))
for T in (0.1, 0.5)
    p = R["V1 atom T=$T upup"][:checks][:C8_parts]
    @printf("F3 — atom T=%.1f (↑↑): max|K^direct| = %.1e, max|K̃| = %.1e, max_p|K_p| = %.2e\n", T, p.maxKd,
            maximum(abs, filter(!isnan, R["V1 atom T=$T upup"][:checks][:box].Kt)), p.maxKp)
end
println("\nF4 — V3c ranges per T (both channels, all Nkeep):")
for T in V3C_TS
    ks = [(T, Nk, c) for Nk in V3C_NKEEPS for c in (:updn, :upup)]
    rng(f) = (minimum(f(v3(k...)) for k in ks), maximum(f(v3(k...)) for k in ks))
    @printf("  T=%.0e: C1 %.1e–%.1e, C2 %.1e–%.1e, C7 %.1e–%.1e, C8 %.1e–%.1e, C3 %.1e–%.1e, C4 %.1e–%.1e, C5 %.1e–%.1e, C6 %.1e–%.1e\n", T,
            rng(o -> o[:checks][:C1])..., rng(o -> o[:checks][:C2])..., rng(o -> o[:checks][:C7])..., rng(o -> o[:checks][:C8])...,
            rng(o -> o[:checks][:C3])..., rng(o -> o[:checks][:C4])..., rng(o -> o[:checks][:C5])..., rng(o -> o[:checks][:C6])...)
end
```

F1 — unstabilized (diagnostic) vs default (stabilized) particle chain, exact chain data

case	C4 unst	C4 big-uns	C4 default	L uns	L def	dimΩ	C1 uns	C8 uns	n(>1e-9)
V2 Nch=0 B=0.0 updn exact	1.03e+00	1.03e+00	1.2e-15	15	14	2	1.5e-15	4.7e-15	15
V2 Nch=1 B=0.0 updn exact	8.45e-01	8.35e-01	1.2e-14	205	203	3	2.0e-15	6.3e-15	133
V2 Nch=1 B=0.0 upup exact	8.30e-01	8.14e-01	1.2e-14	205	203	3	1.8e-15	4.2e-14	133
V2 Nch=1 B=0.05 updn exact	4.74e-01	5.56e-01	5.0e-16	281	279	3	2.4e-15	3.1e-15	183
V2 Nch=1 B=0.05 upup exact	5.50e-01	5.55e-01	5.0e-16	281	279	3	1.4e-15	2.2e-14	189
V2 Nch=2 B=0.0 updn exact	2.48e-01	—	4.9e-15	2615	2612	4	4.2e-15	4.8e-14	1150
V2 Nch=2 B=0.0 upup exact	2.36e-01	—	4.9e-15	2615	2612	4	4.4e-15	2.8e-13	1150
V2 Nch=2 B=0.05 updn exact	1.99e-01	—	5.6e-15	3839	3836	4	6.9e-15	8.4e-14	1626
V2 Nch=2 B=0.05 upup exact	2.01e-01	—	5.6e-15	3839	3836	4	7.4e-15	7.5e-14	1626
V2 block N=2	6.44e-01	—	1.1e-15	281	279	6	2.6e-15	8.8e-15	179

F2 — Steps 1 cases: C2 invariants max 1.2e-12; n_agree / L from 0.08 to 1.00 (n_agree 1–1053)
F3 — atom T=0.1 (↑↑): max|K^direct| = 1.8e-16, max|K̄| = 2.0e-16, max_p|K_p| = 1.09e+00
F3 — atom T=0.5 (↑↑): max|K^direct| = 3.7e-18, max|K̄| = 7.0e-18, max_p|K_p| = 4.34e-02

F4 — V3c ranges per T (both channels, all Nkeep):
  T=1e-02: C1 2.2e-06–1.3e-03, C2 3.0e-14–5.8e-13, C7 2.4e-08–2.1e-02, C8 8.9e-05–1.2e-01, C3 2.5e-01–5.6e-01, C4 4.2e-01–5.8e-01, C5 0.0e+00–0.0e+00, C6 2.3e-16–2.1e-15
  T=1e-04: C1 3.4e-05–1.4e-02, C2 2.2e-14–1.3e-13, C7 2.0e-04–2.3e-02, C8 4.5e-04–5.8e-01, C3 2.6e-01–6.0e-01, C4 4.6e-01–6.4e-01, C5 0.0e+00–0.0e+00, C6 6.4e-07–1.4e-05